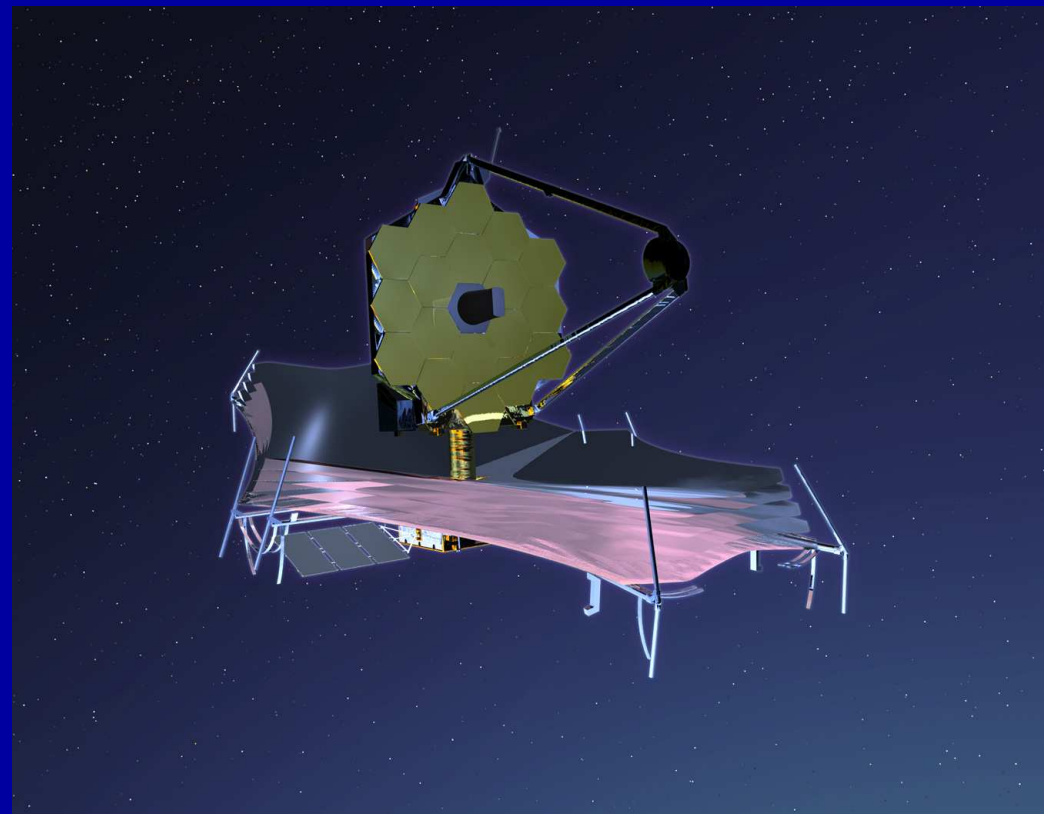


Using Hubble Object Finding and Classification Software to Find Markers for Diabetes Type 2 in an Early Stage

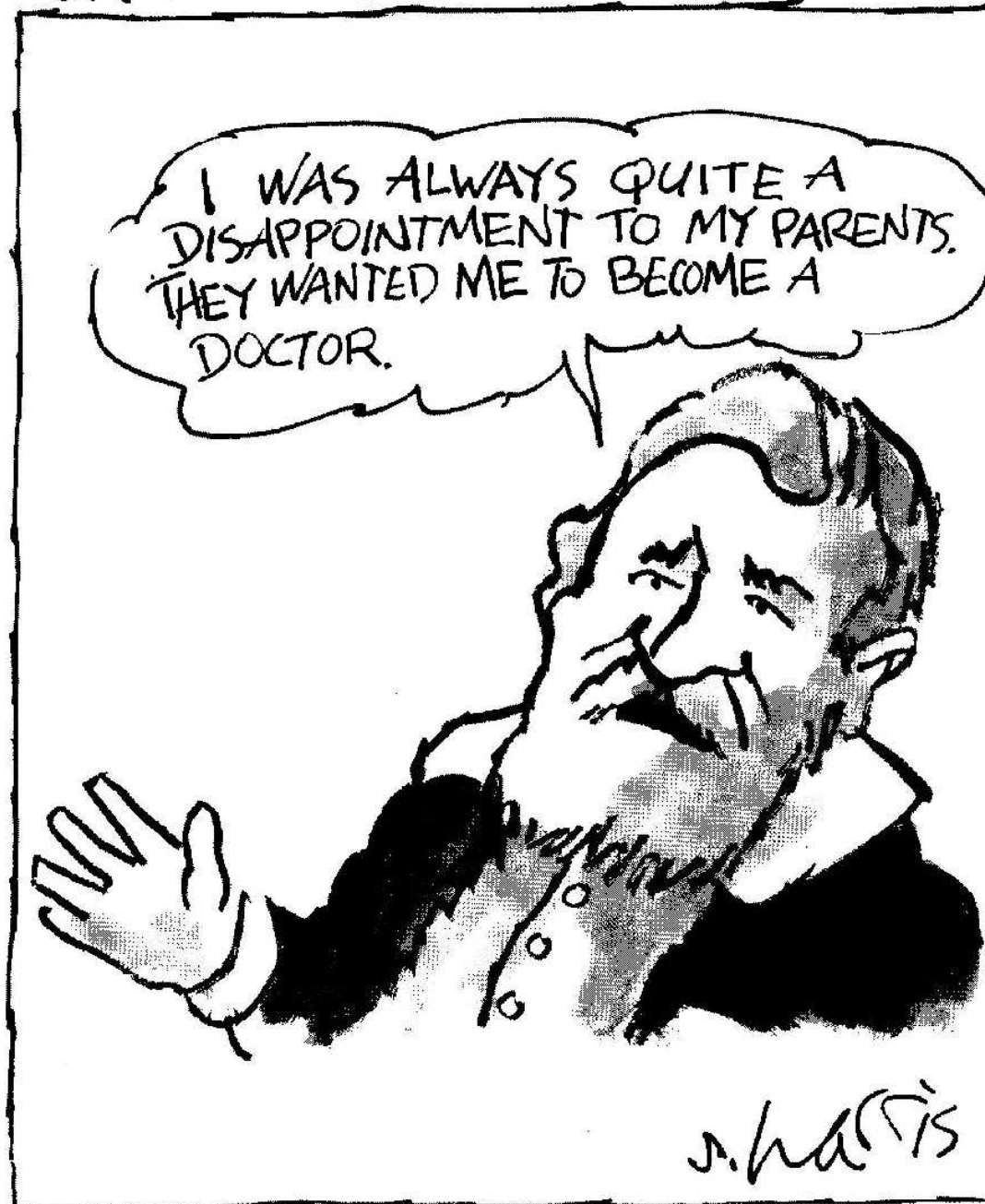
Rogier Windhorst (School of Earth and Space Exploration, ASU)

P. Baluch, L. Burnett, J. Brower, A. Casano, J. Olson (ASU), R. & J. Herman (Banner Health), R. Ryan, K. Tamura, V. Taylor (ASU)



Invited Lecture at the AIMS Conference, ASU, Friday March 12, 2010, Tempe, AZ

GALILEO'S BURDEN



Why I personally got involved in medical research ... (true story!)

Outline

(1) Summary of the NASA Hubble Space Telescope connection

- (1a) NASA missions to study Cosmic Dawn, First Light, Dark Ages
- (1b) NASA Hubble Space Telescope (HST) image analysis

(2) Using Hubble Object Finding and Classification Software to Find Potential Markers for Diabetes Type 2 in an Early Stage

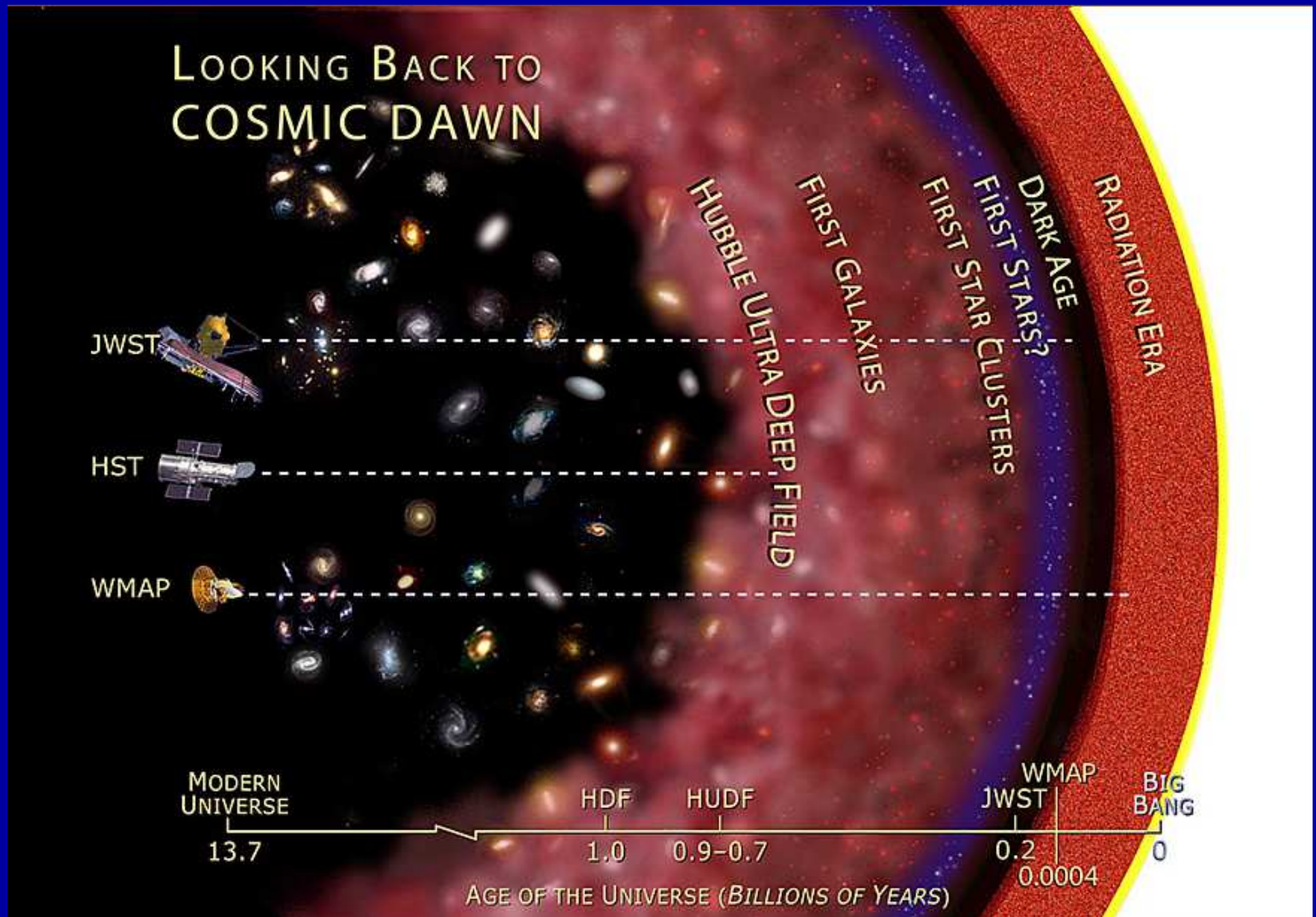
- (2a) The Object Finding, Detection, Deblending, and Classification method
- (2b) Application to C-fiber Skin-biopsies to Normal and Obese Patients

(3) Results

(4) Conclusions

Sponsored by NASA/ASU/Banner Health

(1a) NASA missions that study Cosmic Dawn, First Light, & the Dark Ages



Looking back in time $\Rightarrow$ deal with many overlapping galaxy images!

Intro: Cosmic Expansion and Contents of the Universe

Expansion $\Rightarrow$ redshift

$$\lambda_{obs} = \lambda_{rest} \cdot (1+z)$$

Hubble's Law:

$$D \simeq v / H_0 \simeq (c/H_0) \cdot z = R_0 \cdot z$$

Cosmic Content:

inside $R_0 = (c/H_0) \simeq 13.73$ Glyr:

$$[t_{univ} = (211 \pm 1 \text{ !}) \cdot (t_{dino} = 65 \text{ Myr})]$$

Photons (light):

$$N_{h\nu} \sim 10^{89}$$

Baryons (atoms):

$$N_b \sim 10^{80}$$

η = Photons/Baryons

$$\eta \sim 10^9 \implies \text{He/H ratio} = 0.235$$

Energy Density:

as fraction of critical closure density:

Baryons (atoms):

$$\Omega_b = \rho_b / \rho_{crit} \simeq 0.042$$

Dark Matter:

$$\Omega_d = \rho_d / \rho_{crit} \simeq 0.20$$

Dark Energy (Λ):

$$\Omega_\Lambda = \rho_\Lambda / \rho_{crit} \simeq 0.76$$

(Supermassive) black holes:

$$\rho_{smbh} / \rho_{crit} \simeq 0.0001$$

Total

$$\Omega_{tot} = \rho_{tot} / \rho_{crit} \simeq 1.00 \pm 0.02$$



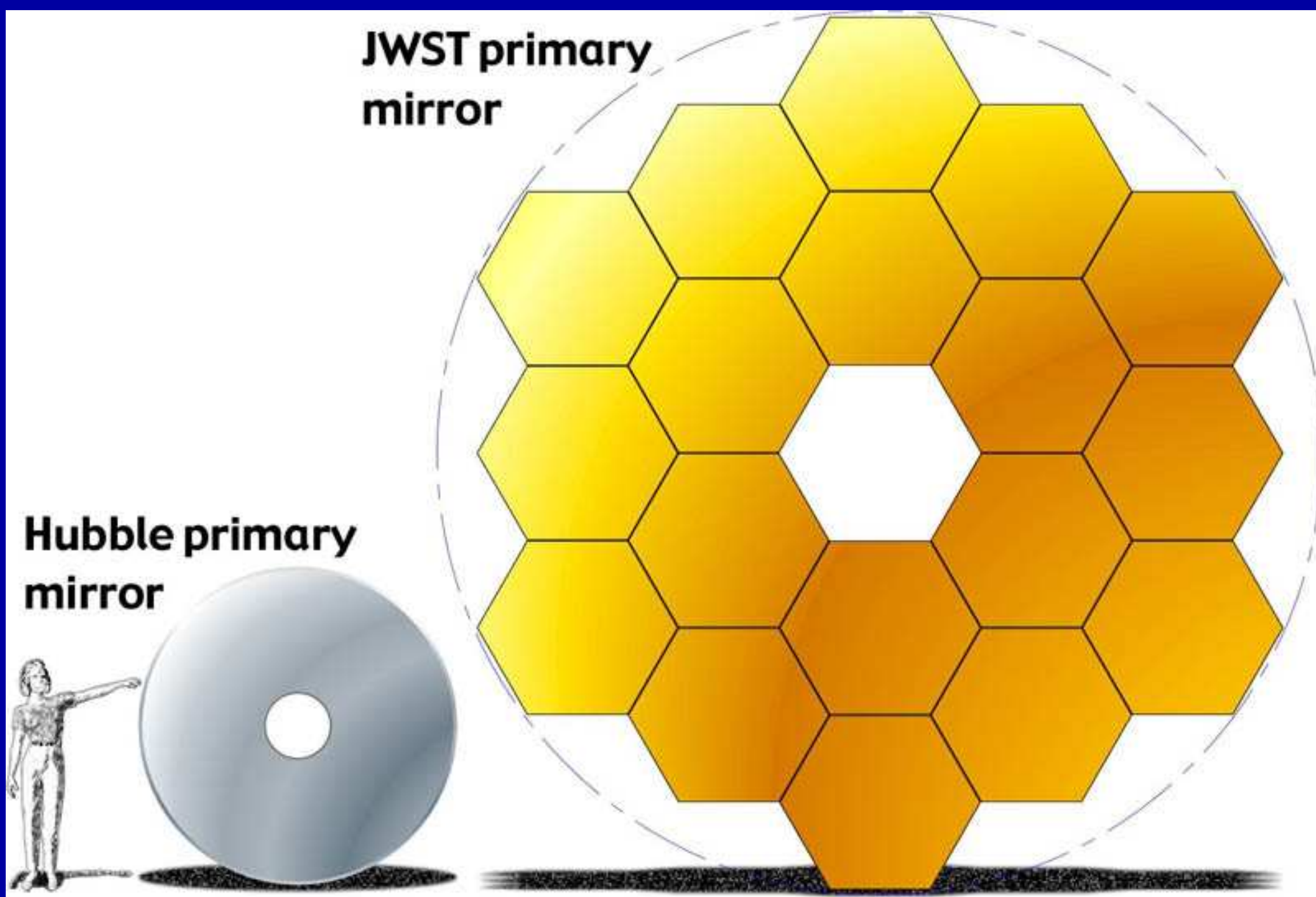
Edwin P. Hubble (1889–1953) — Carnegie astronomer



James E. Webb (1906–1992) — Second NASA Administrator

HST: Concept in 1970's; Made in 1980's; Operational 1990– $\gtrsim$ 2014

JWST: The infrared sequel to HST from 2014–2019 (–2025?)



JWST $\sim 2.5\times$ larger than Hubble, so at $\sim 2.5\times$ larger wavelengths:
JWST has the same resolution in the near-IR as HST in the optical.

(1a) Recent key aspects of the Hubble Space Telescope (HST) project



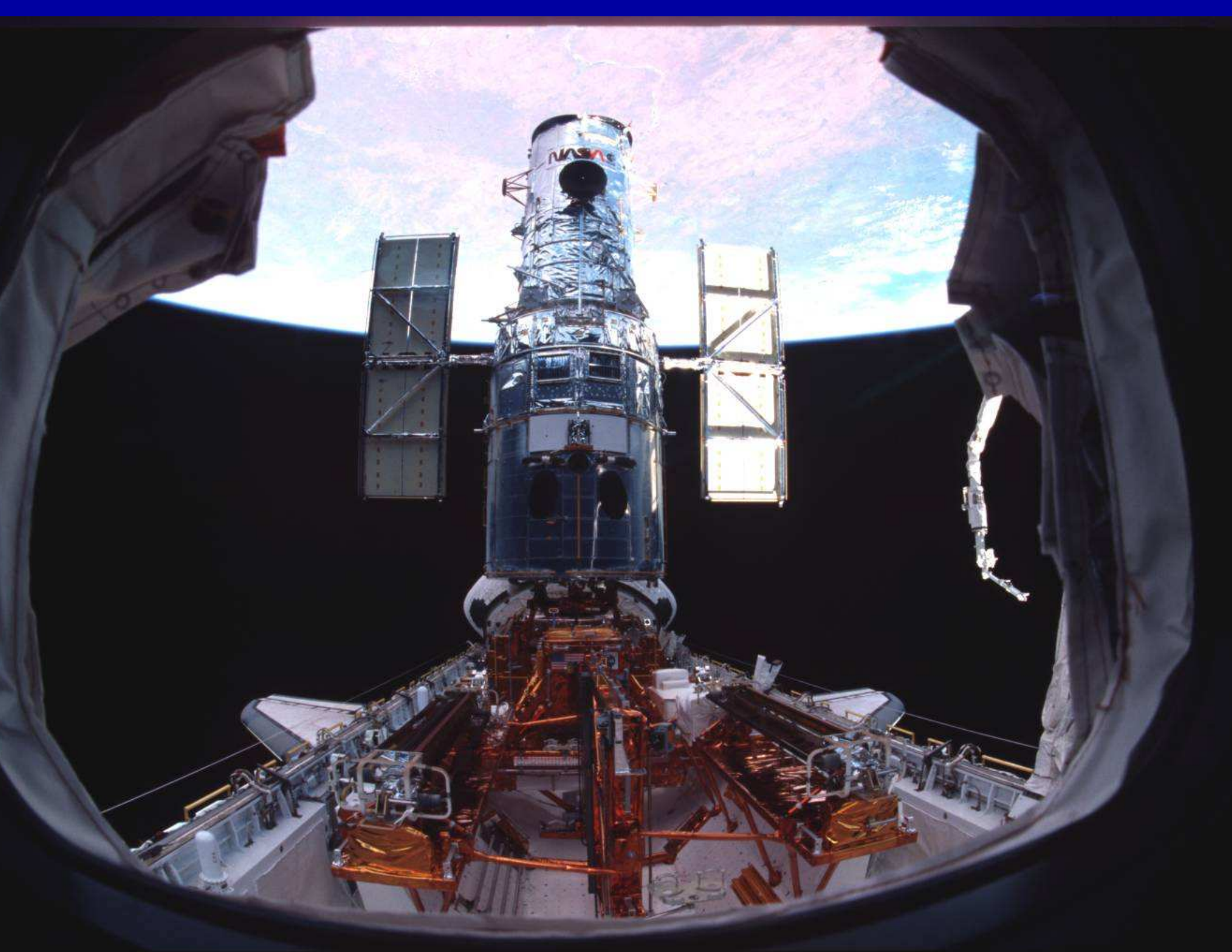
The HST Advanced Camera for Surveys (ACS) — launched 2002 (SM3B).



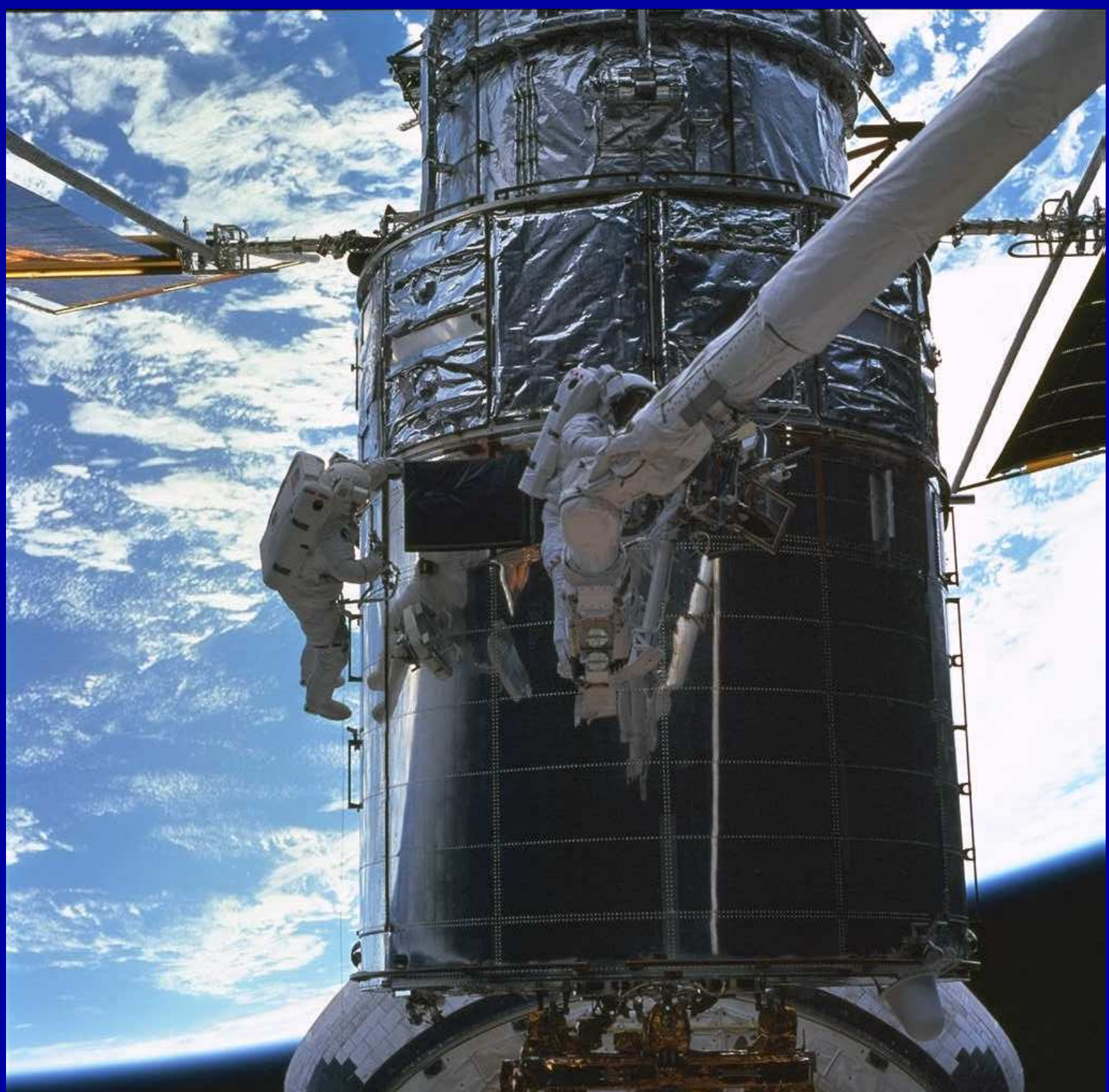


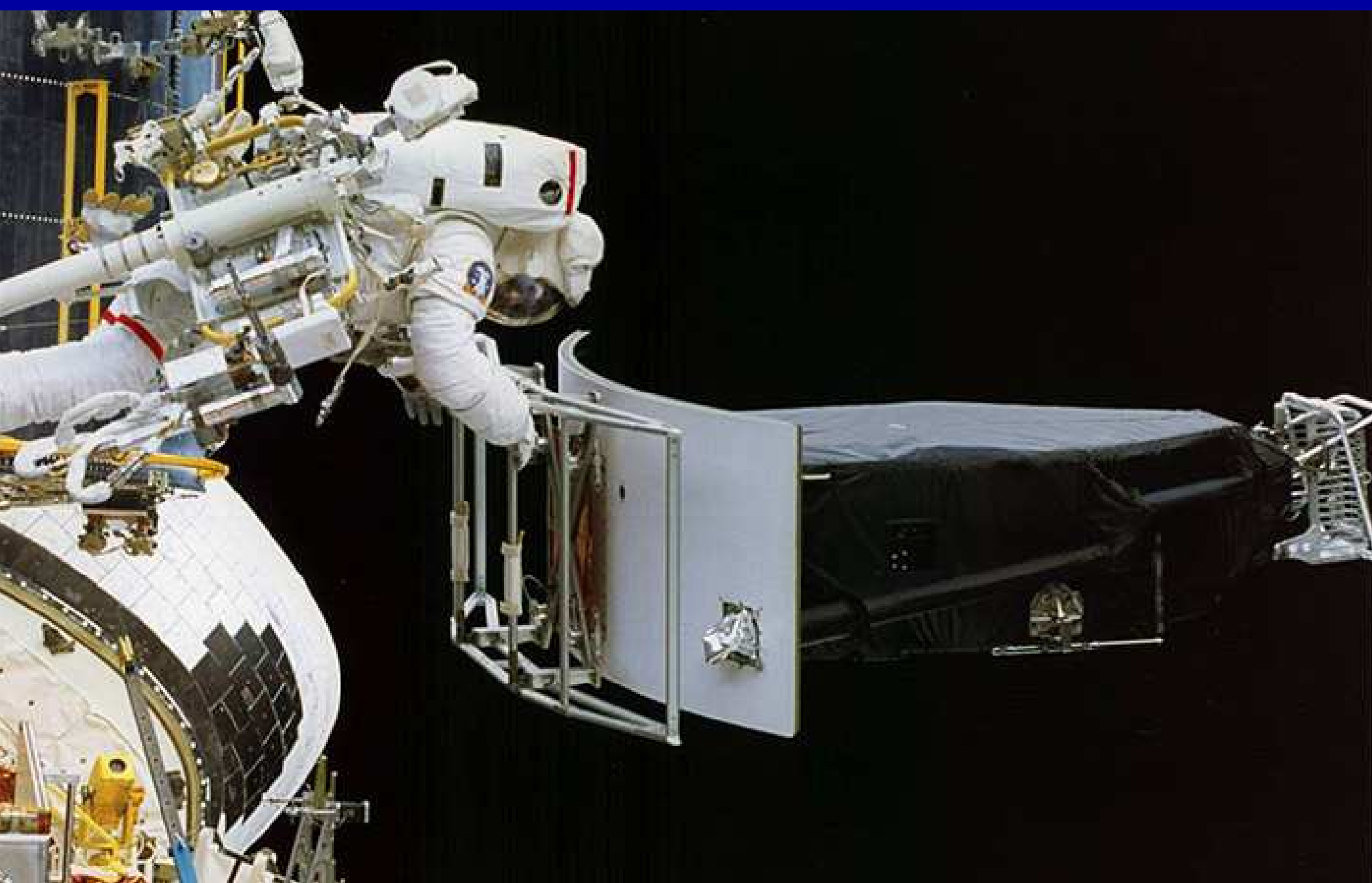




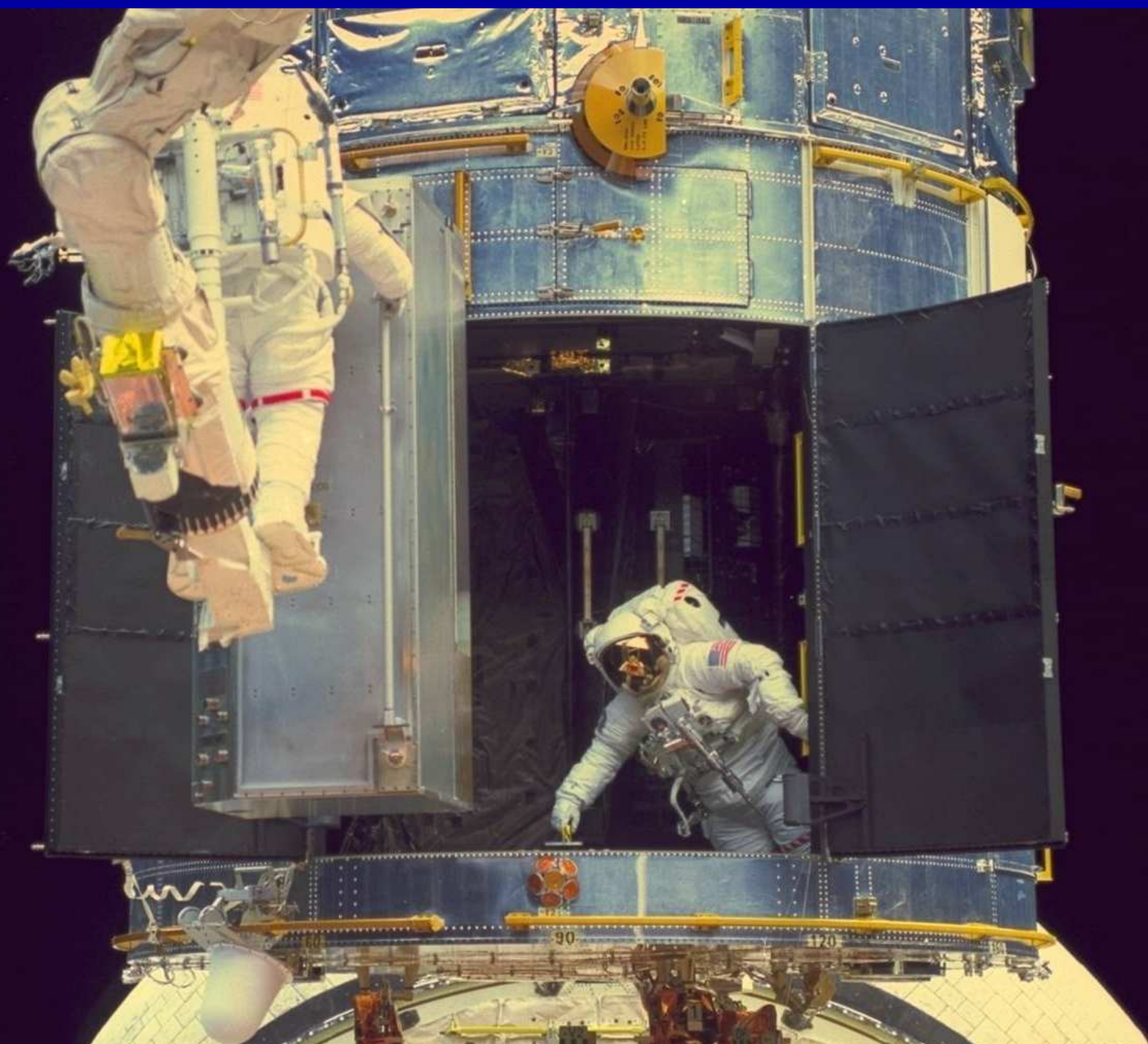








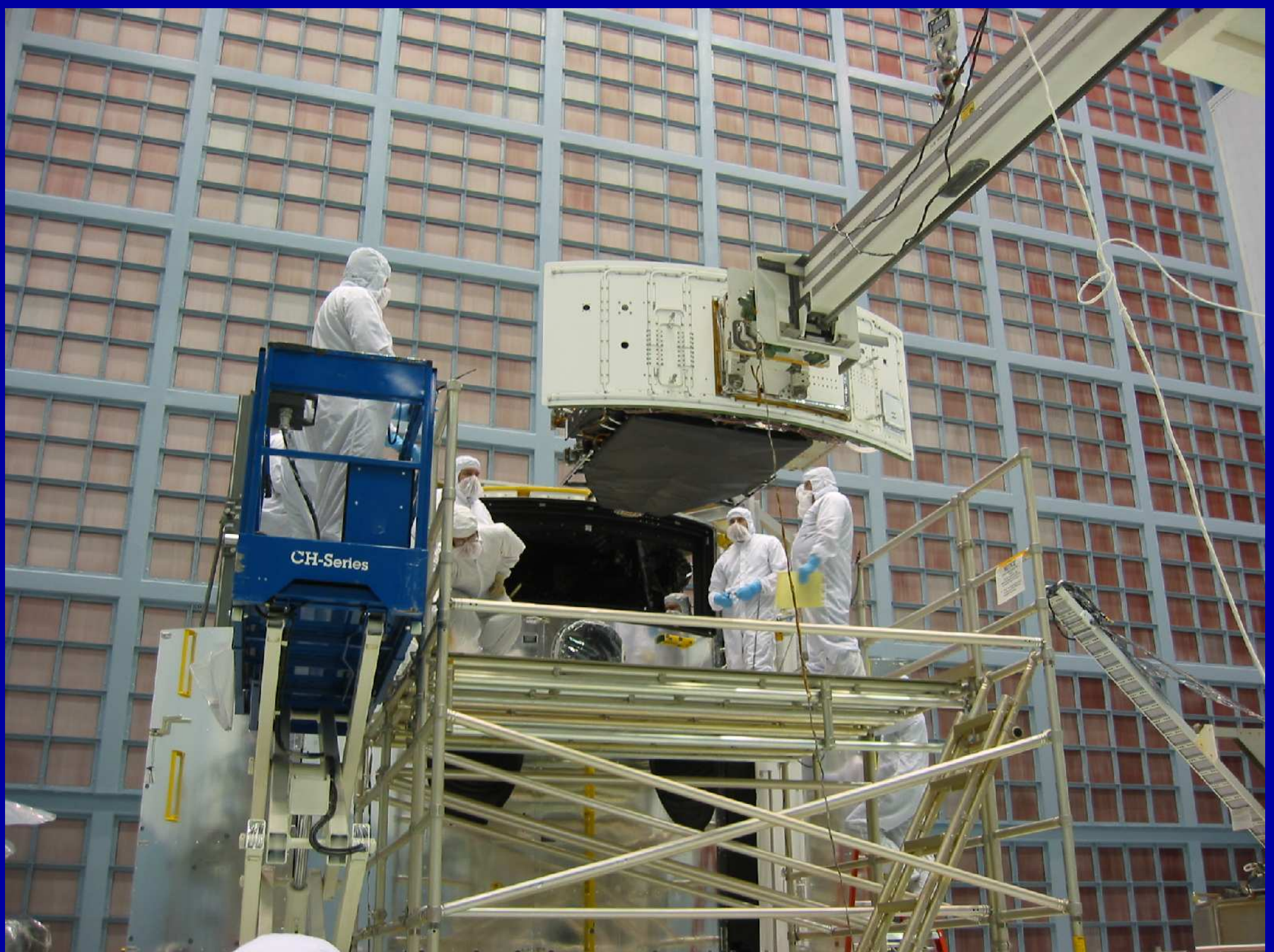
Installing Wide Field Camera 2 (WFPC2) during SM1 in December 1993.
Similar to what astronauts did with WFC3 during SM4 in May 2009.





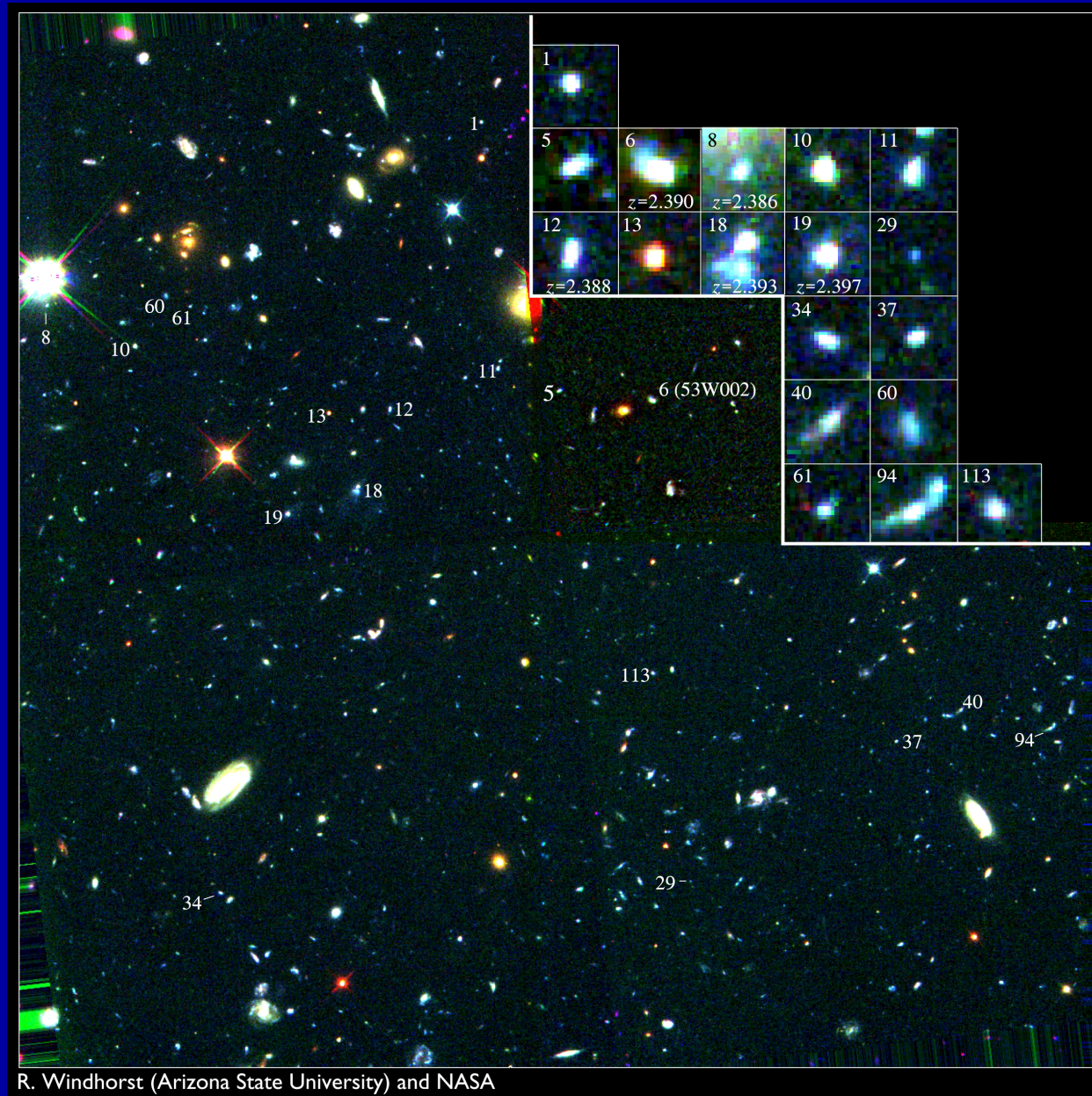
New ESA solar panels rolling out during SM1 in December 1993





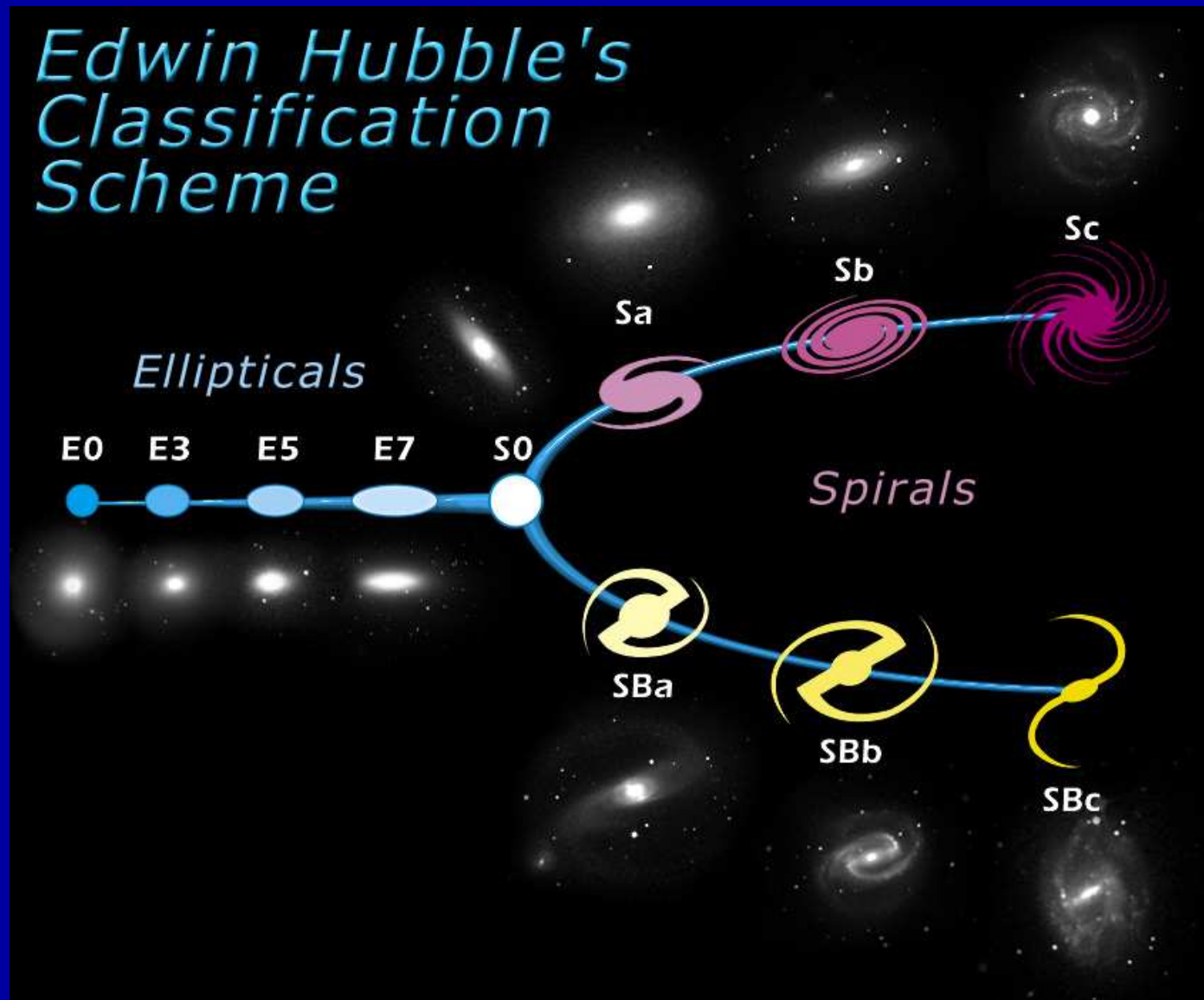
Wide Field Camera 3 for SM4 in 2009: More powerful HST imaging than ever.

(1a) How has HST measured Galaxy Assembly over Cosmic Time?



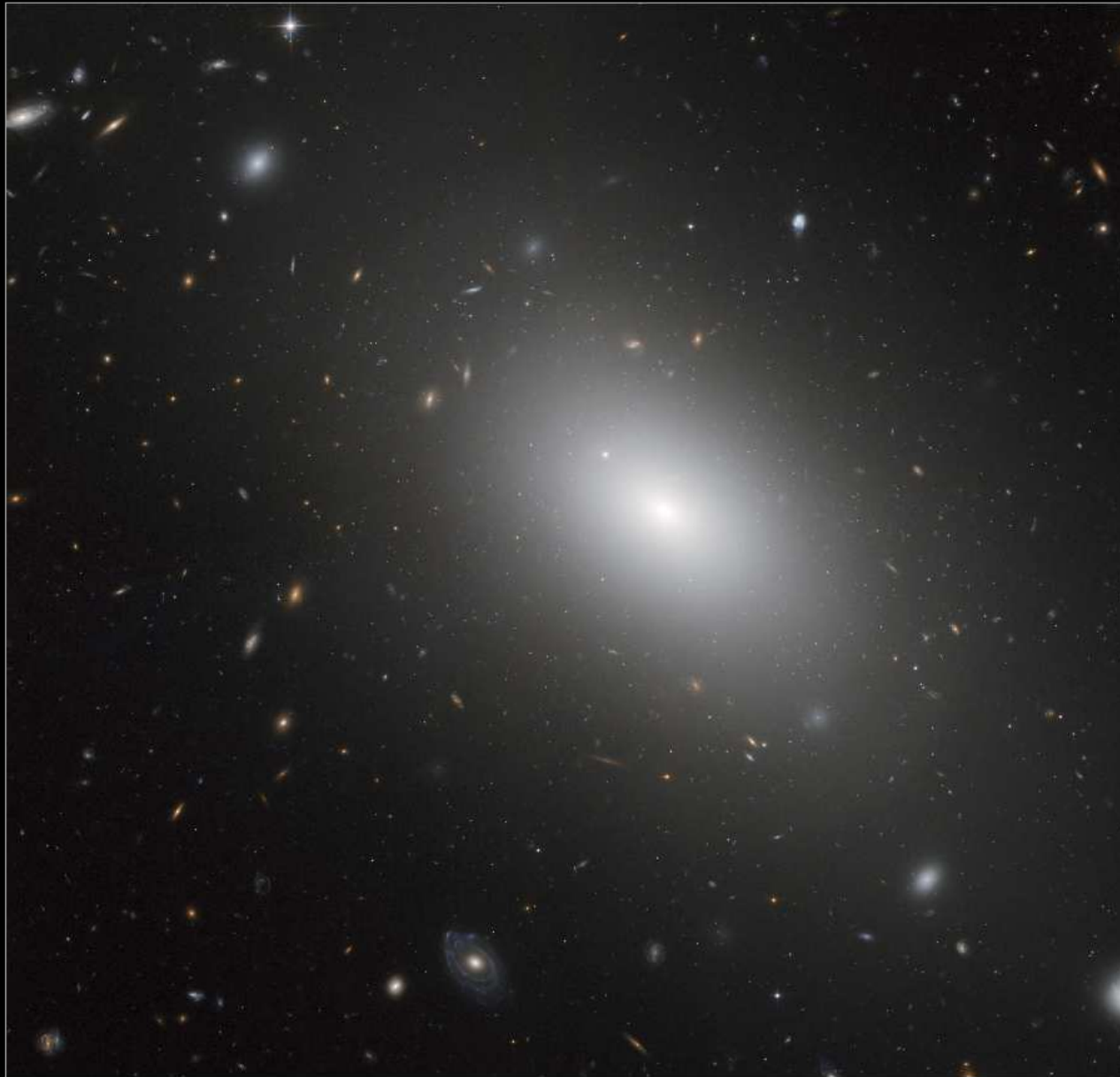
One of the remarkable discoveries of HST was how numerous and small faint galaxies are — the building blocks of the giant galaxies seen today.

HST turned the classical Hubble sequence upside down!

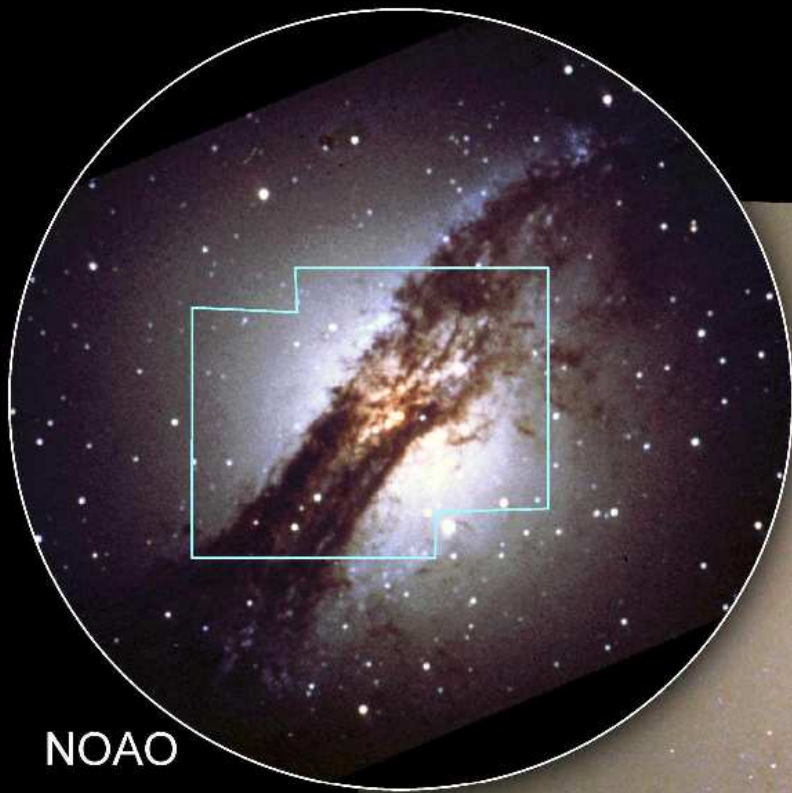


Who (when)	Cosmic Epoch	Ellipticals	Spirals	Irr's/mergers
Hubble (1920's)	$z=0$ (13.73 Gyr)	$\sim 40\%$	$\gtrsim 50\%$	$\lesssim 10\%$
HST (1990's)	$z \simeq 1-2$ (3-6 Gyr)	$\lesssim 15\%$	$\sim 30\%$	$\gtrsim 55\% !$

Elliptical Galaxy NGC 1132

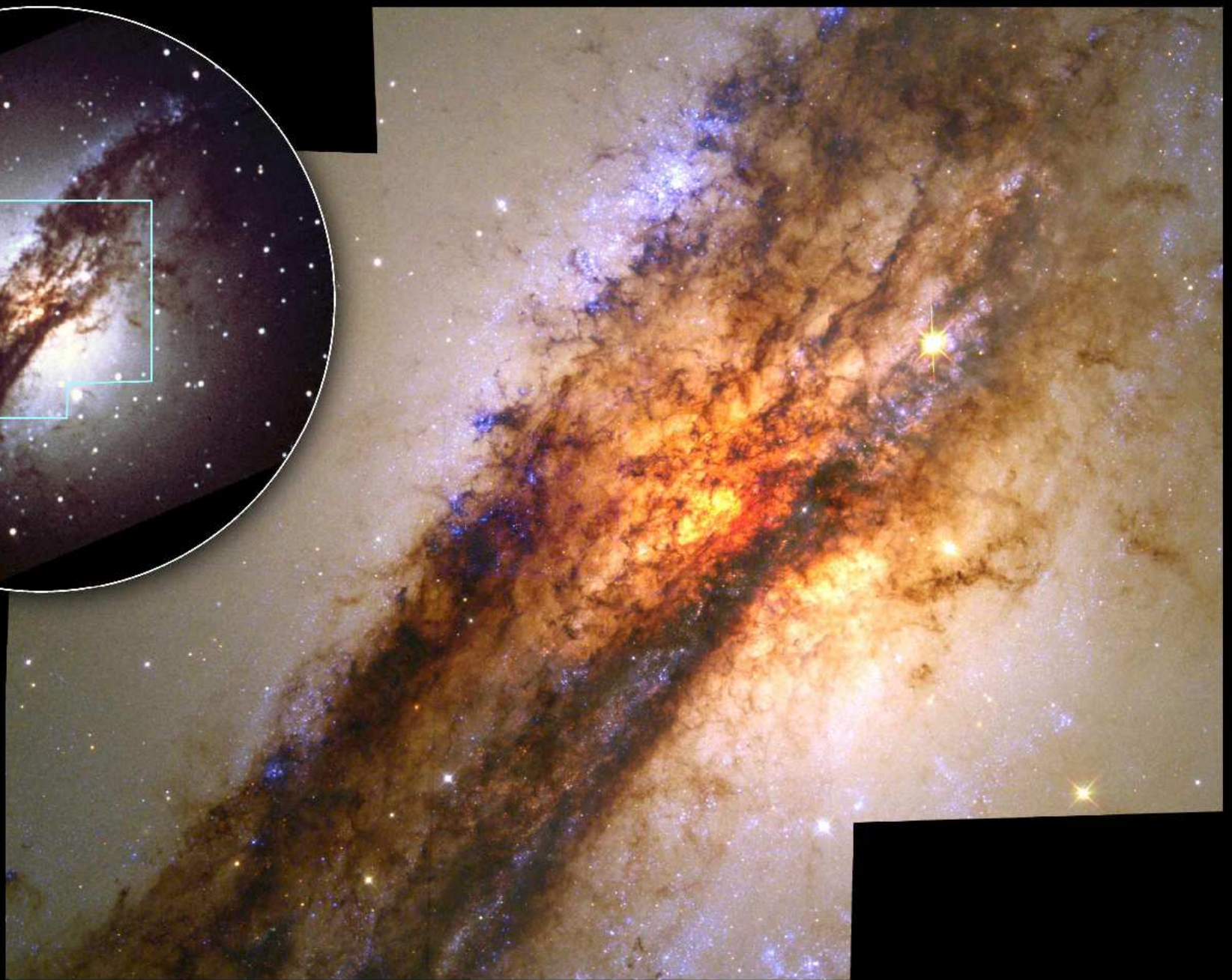


Hubble
Heritage



NOAO

HST



Active Galaxy Centaurus A

Hubble Space Telescope • Wide Field Planetary Camera 2

Elliptical Galaxy NGC 1316



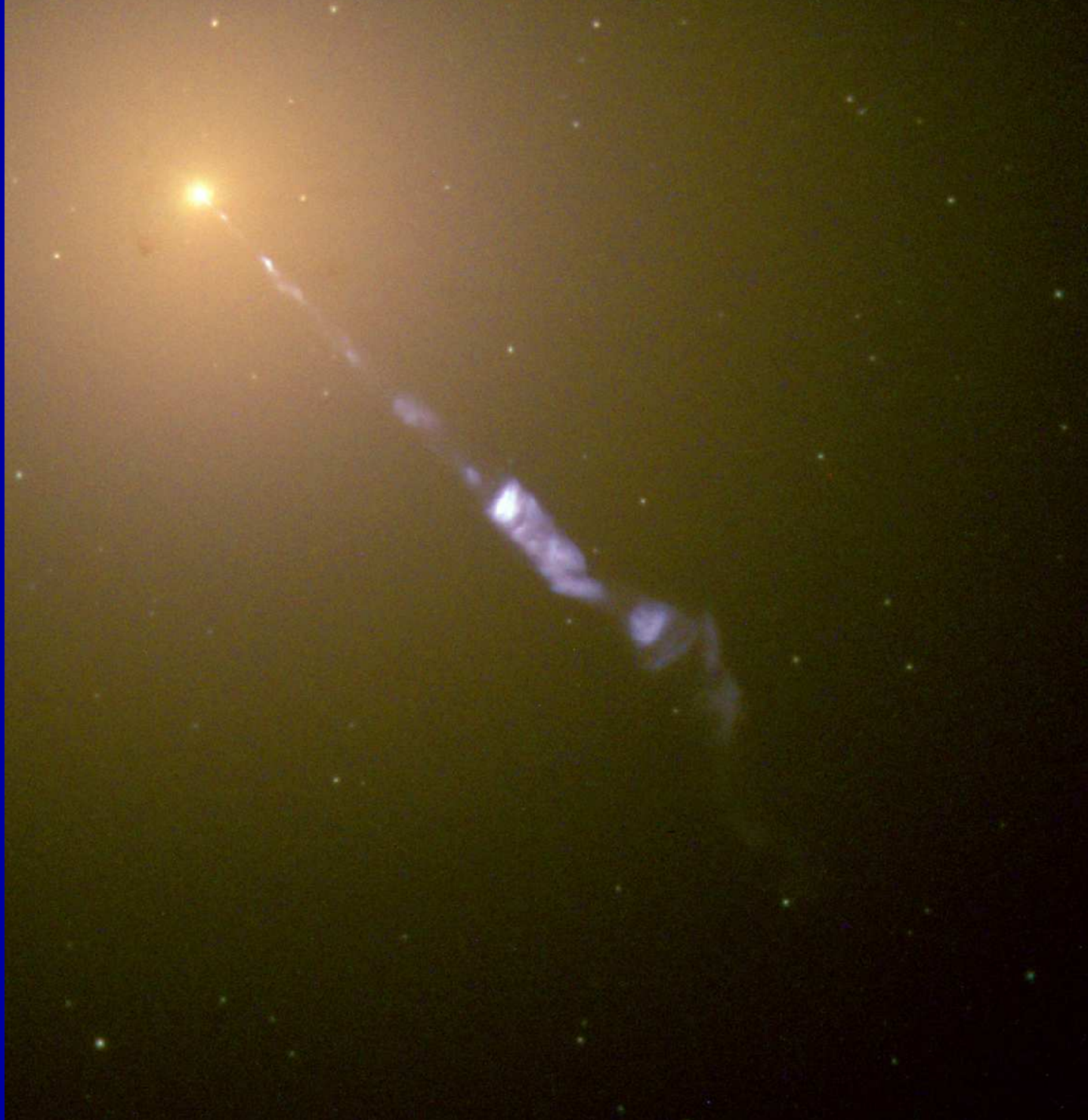


Active Galaxy M82



Hubble
Heritage

NASA, ESA, and The Hubble Heritage Team (STScI/AURA) • Hubble Space Telescope ACS/WFC • STScI-PRC06-14a



Elliptical galaxy M87 with Active Galactic Nucleus (AGN) and relativistic jet.

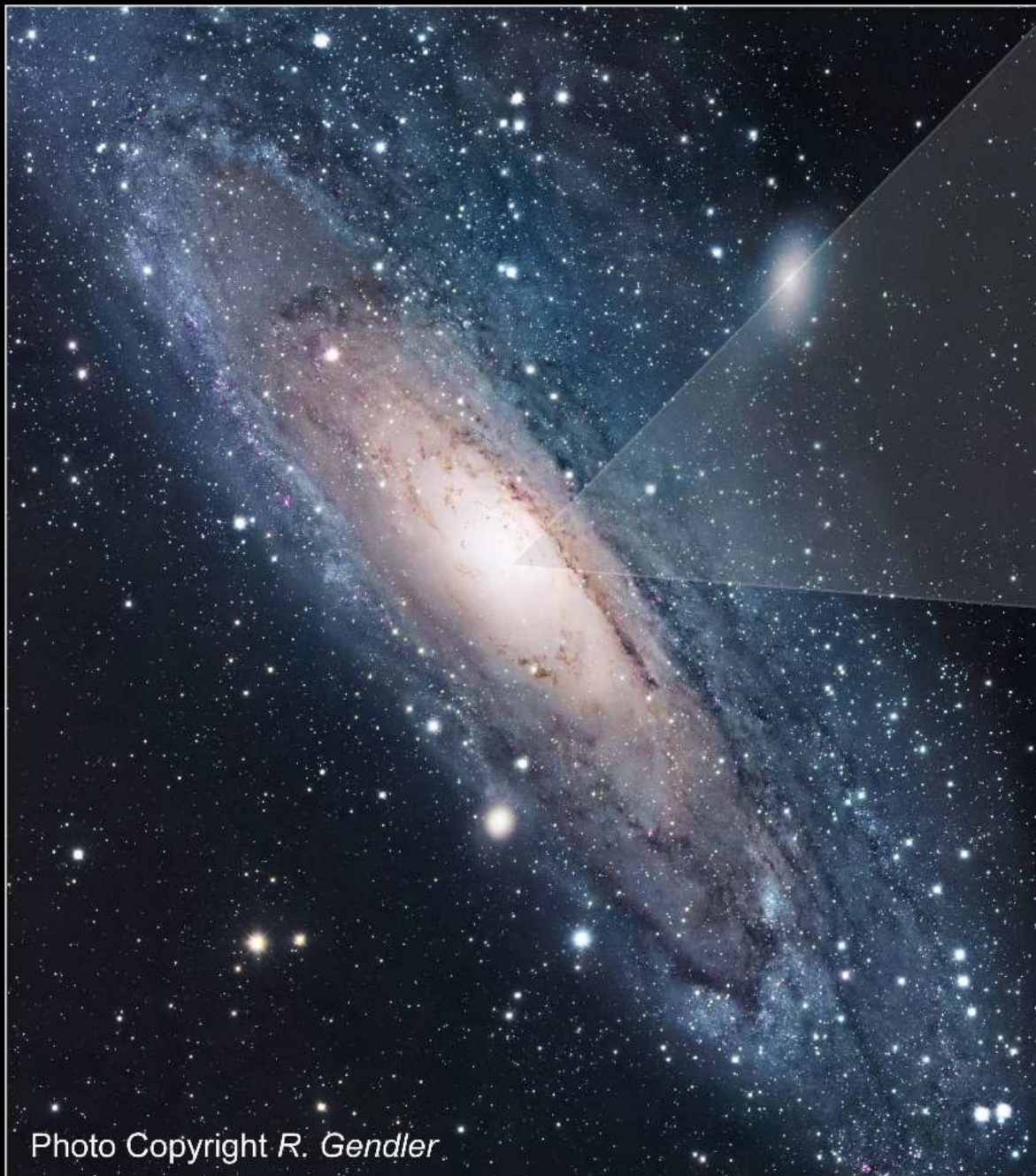
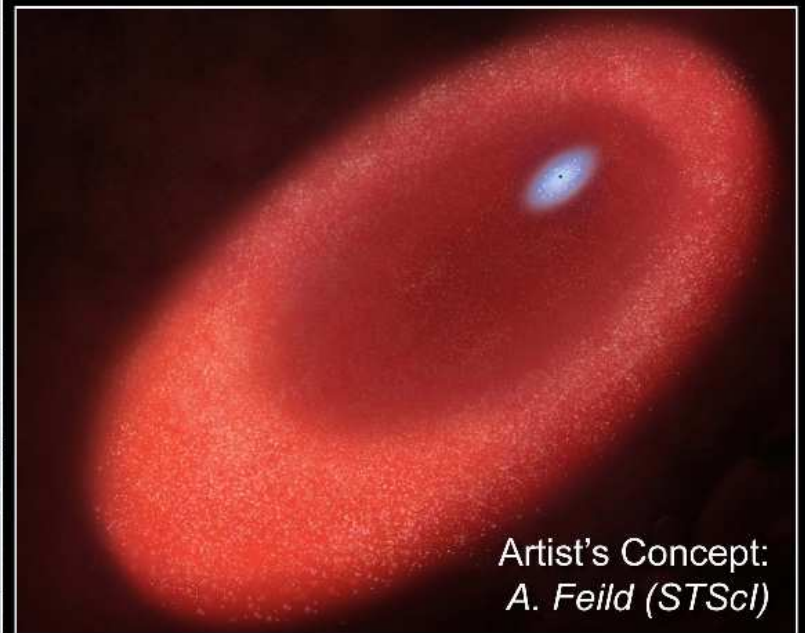


Photo Copyright *R. Gendler*



*HST WFPC2 image:
T. Lauer (NOAO/AURA/NSF)*



*Artist's Concept:
A. Feild (STScI)*

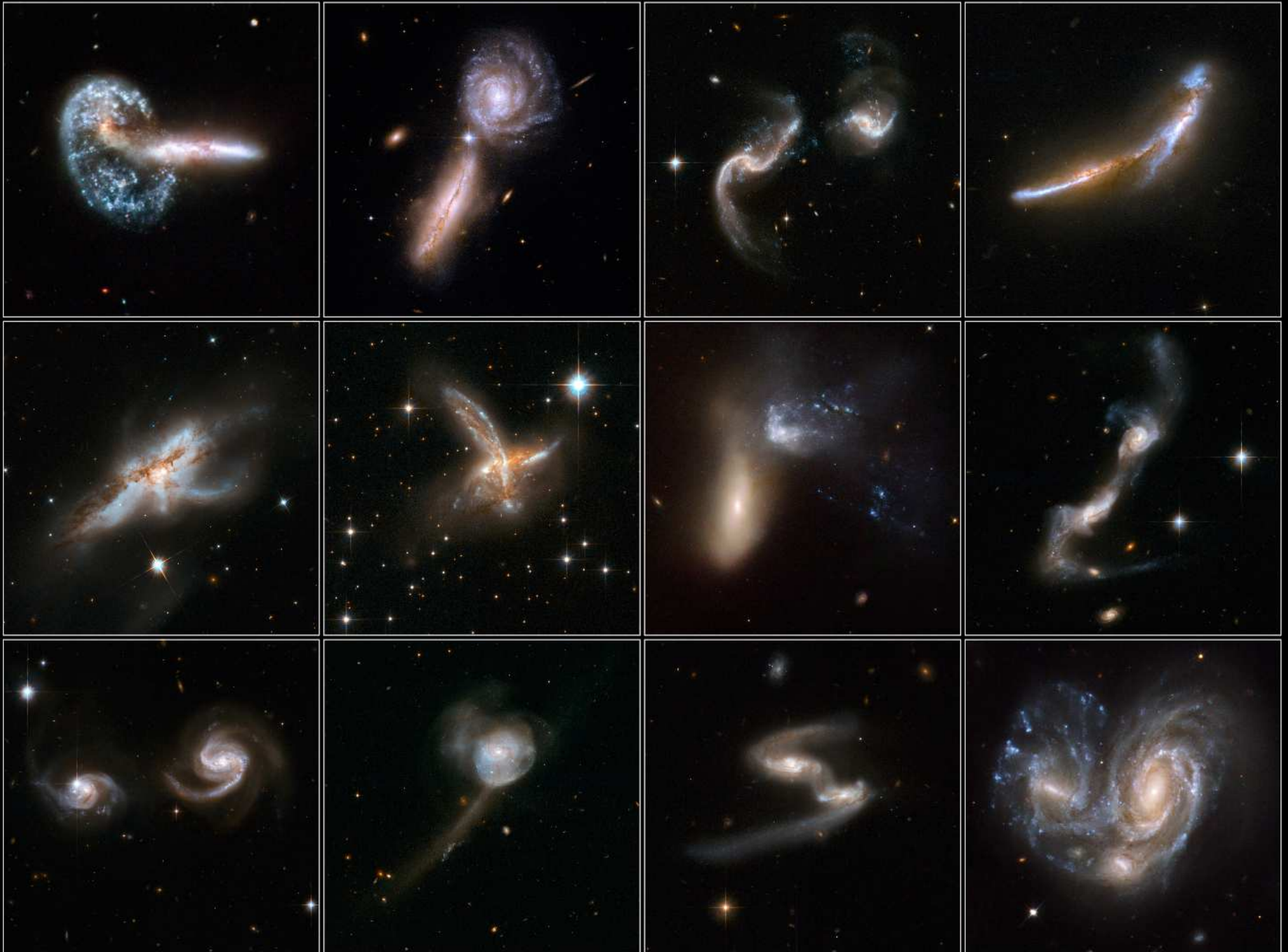
Andromeda Galaxy Nucleus ■ M31
Hubble Space Telescope ■ WFPC2

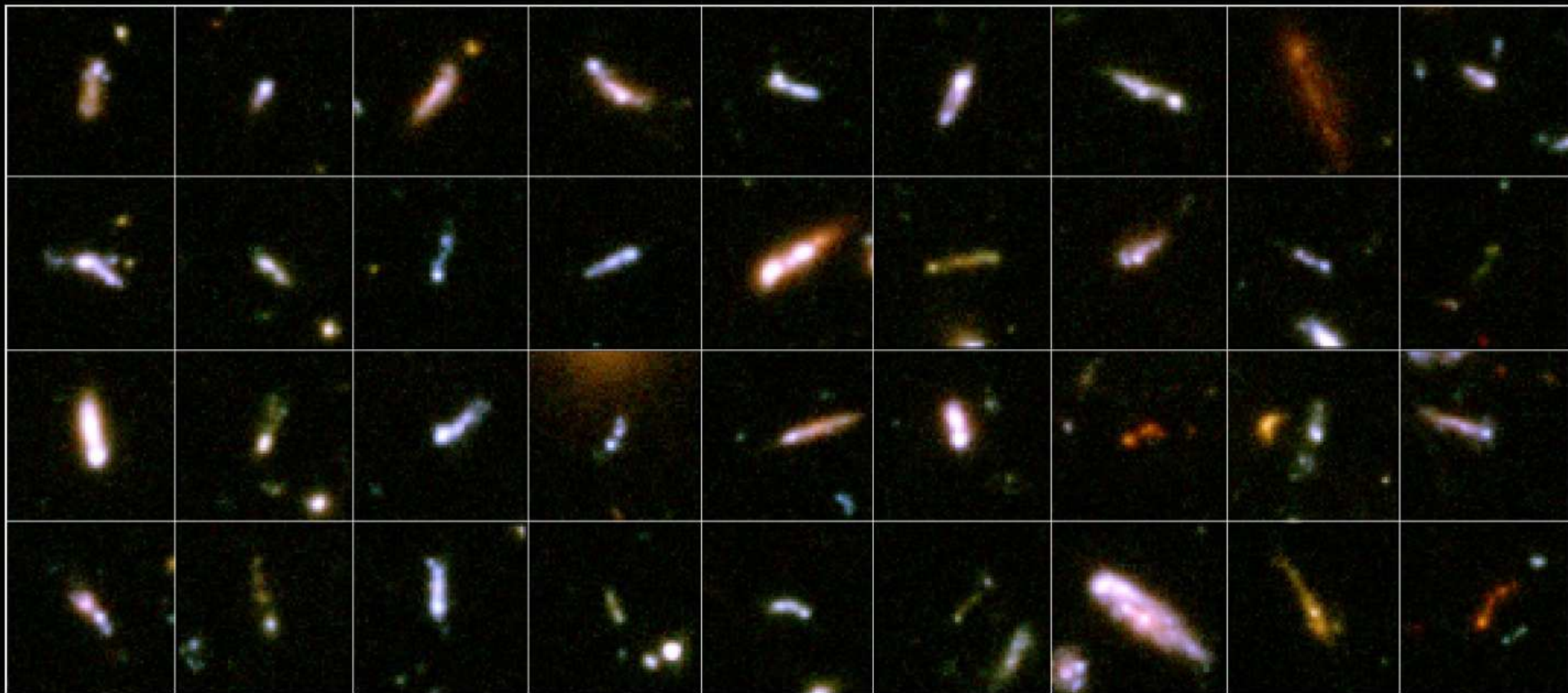


HST Antenna galaxy: Prototype of high redshift, star-forming, major merger?

Interacting Galaxies

Hubble Space Telescope • ACS/WFC • WFPC2





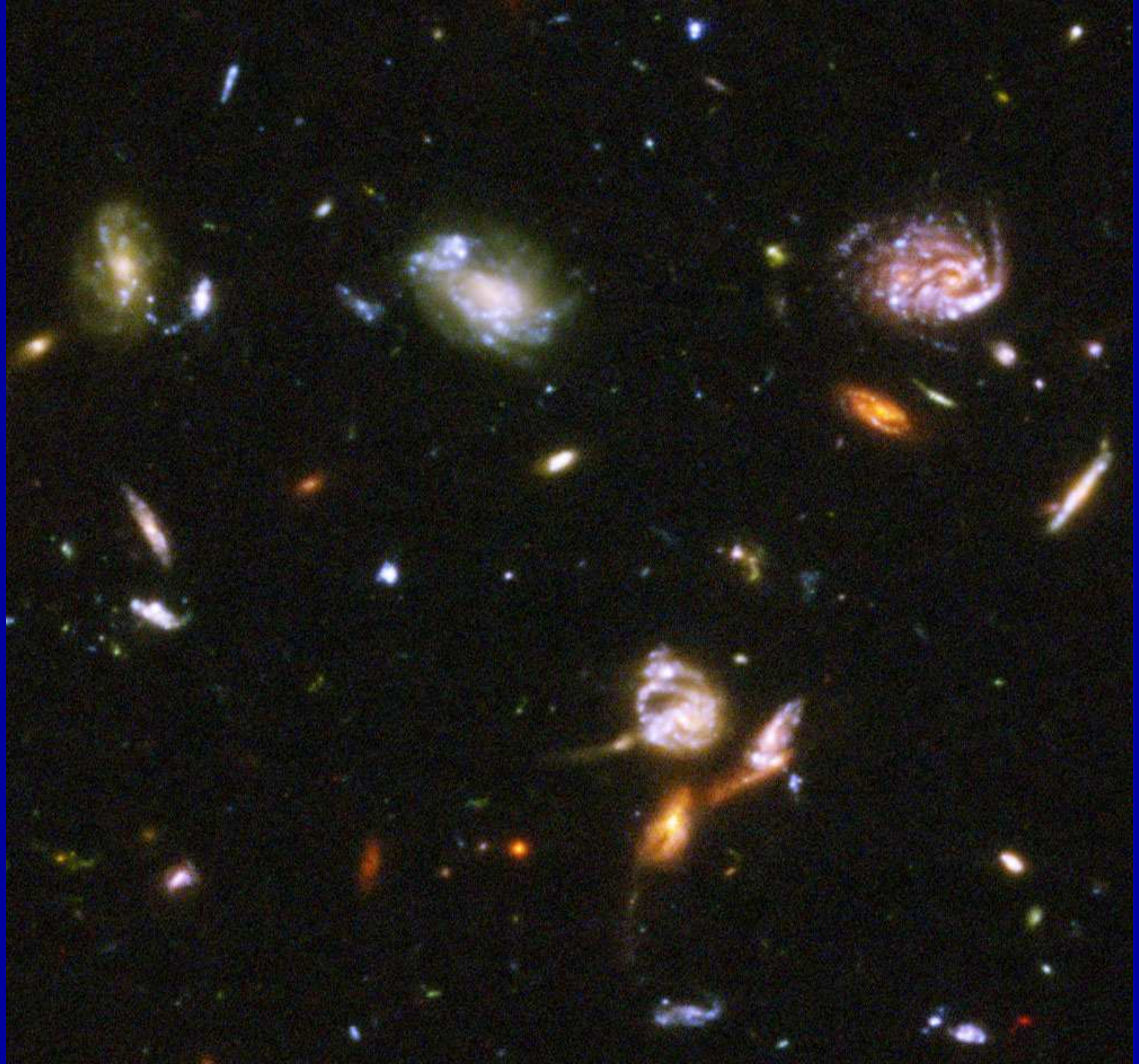
“Tadpole” Galaxies in the Hubble Ultra Deep Field
Hubble Space Telescope ■ ACS/WFC

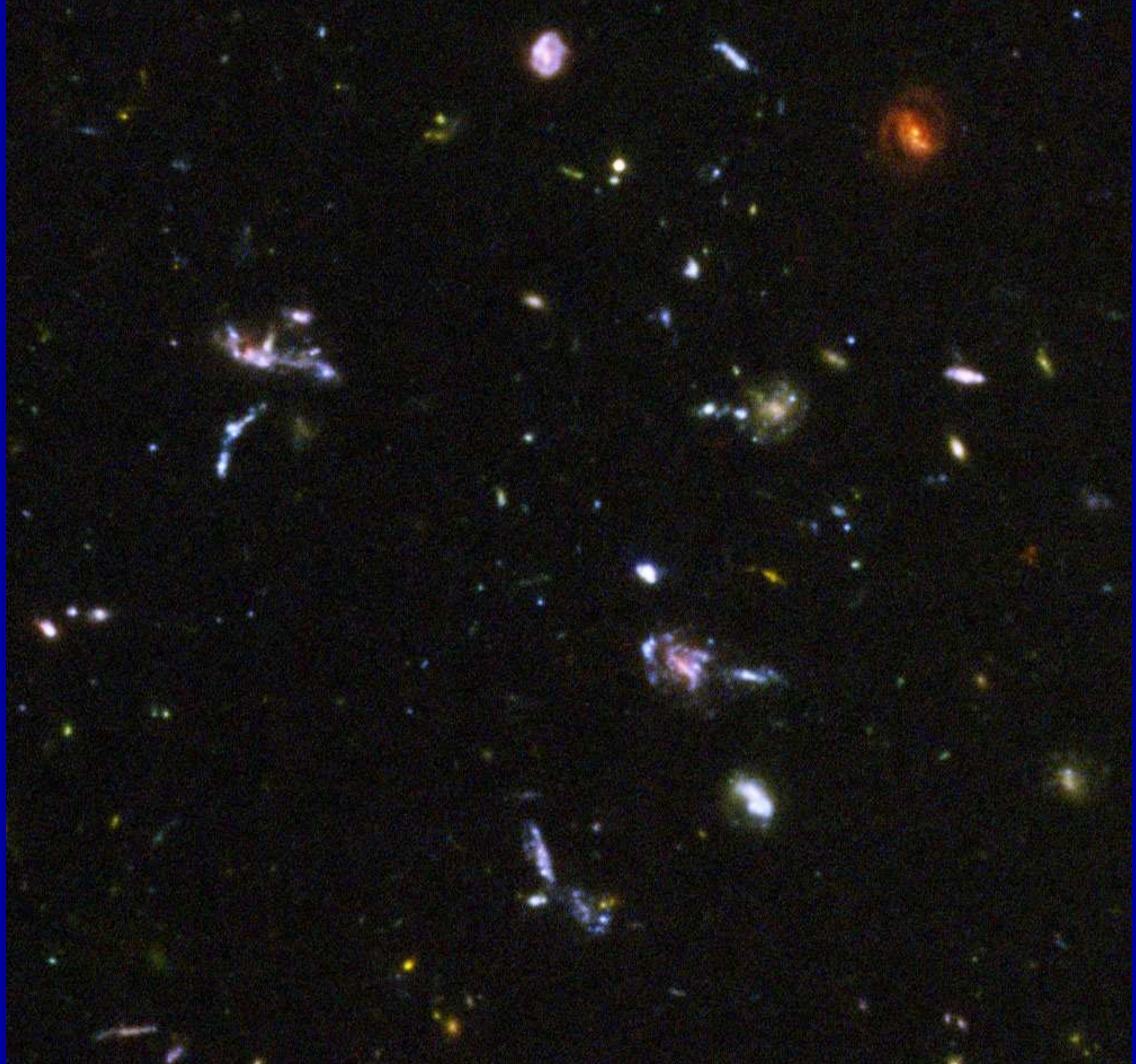
NASA, ESA, A. Straughn, S. Cohen and R. Windhorst (Arizona State University), and the HUDF team (STScI)

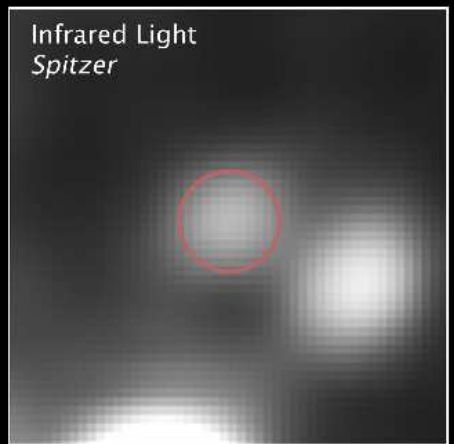
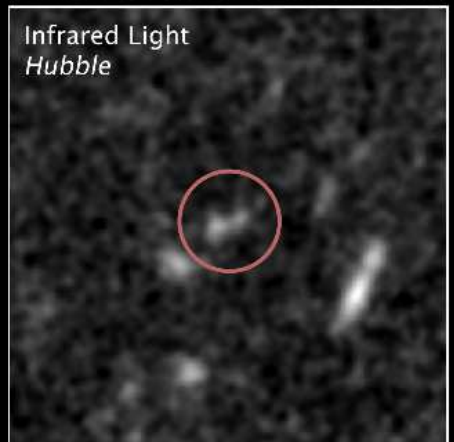
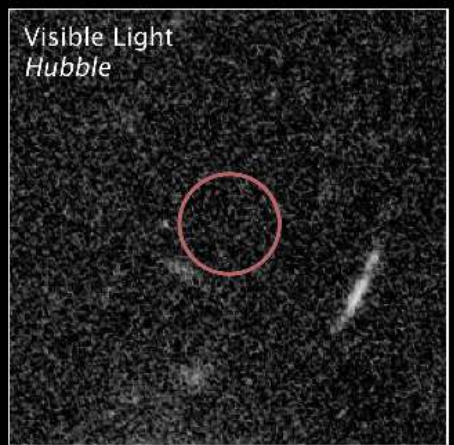
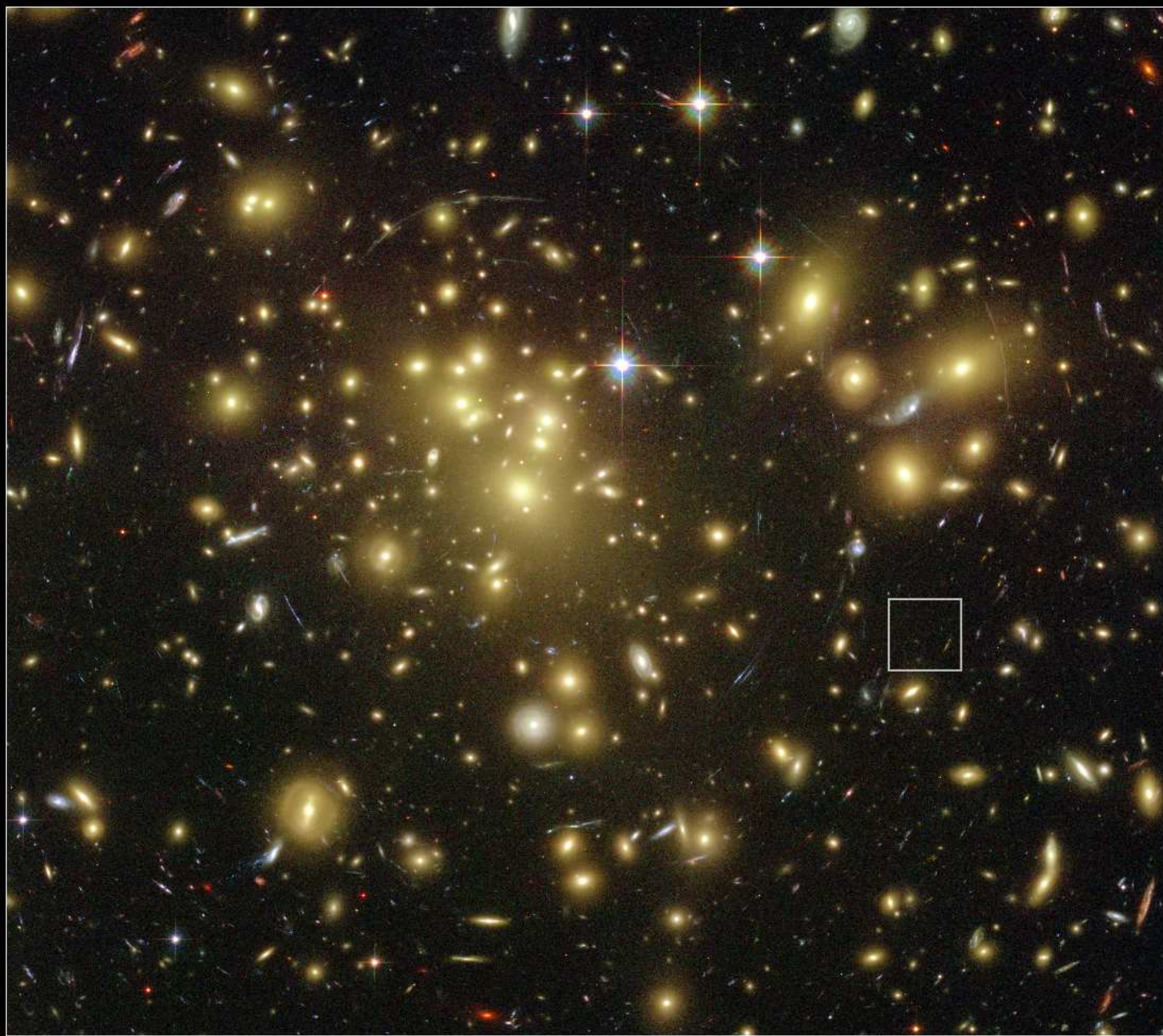
STScI-PRC06-04

Merging galaxies constitute $\lesssim 1\%$ of Hubble sequence today (age $\gtrsim 12.5$ Gyr).

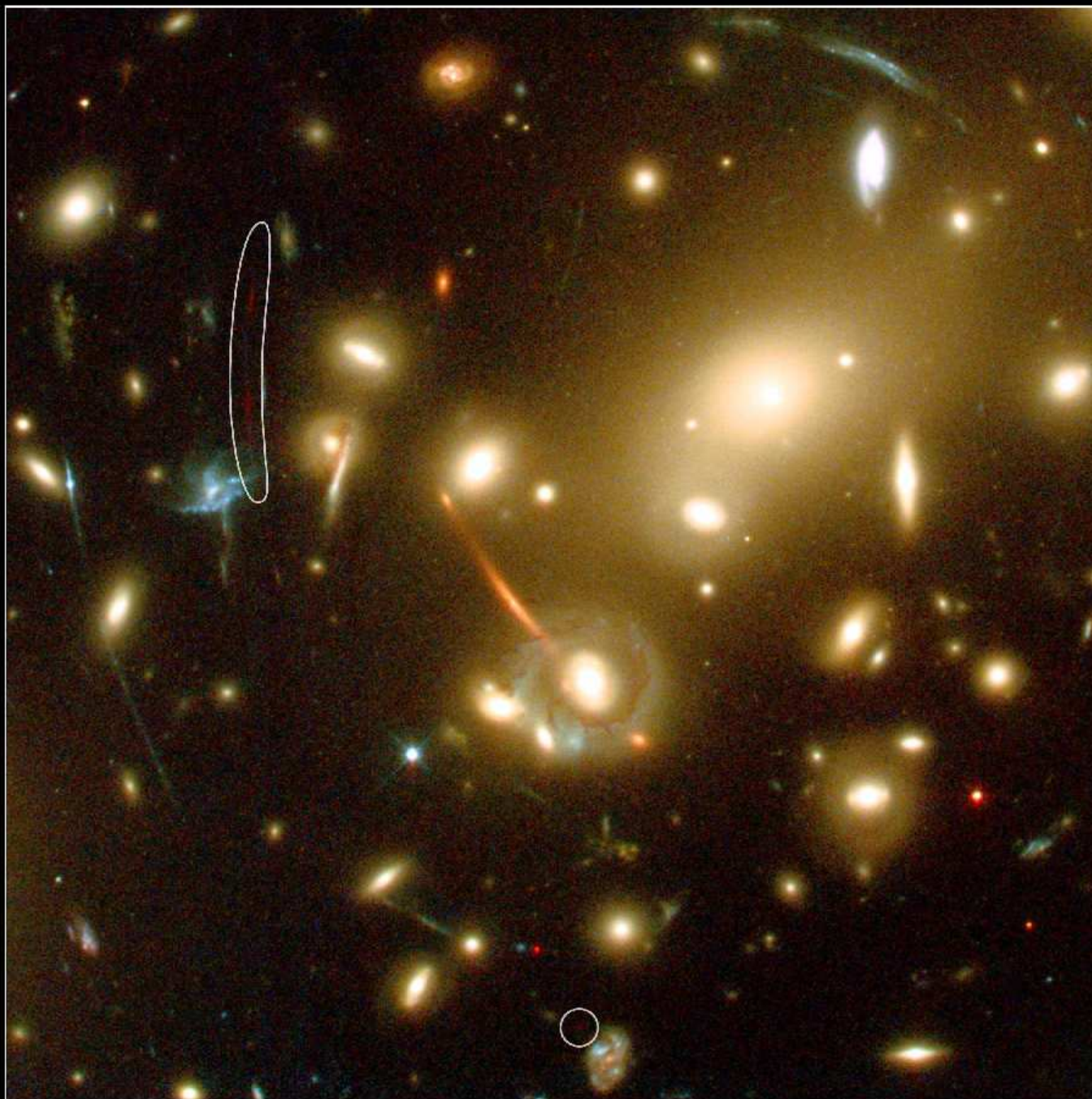
Tadpole galaxies are early stage mergers, very common at $z \gtrsim 2$ (age $\lesssim 3$ Gyr).



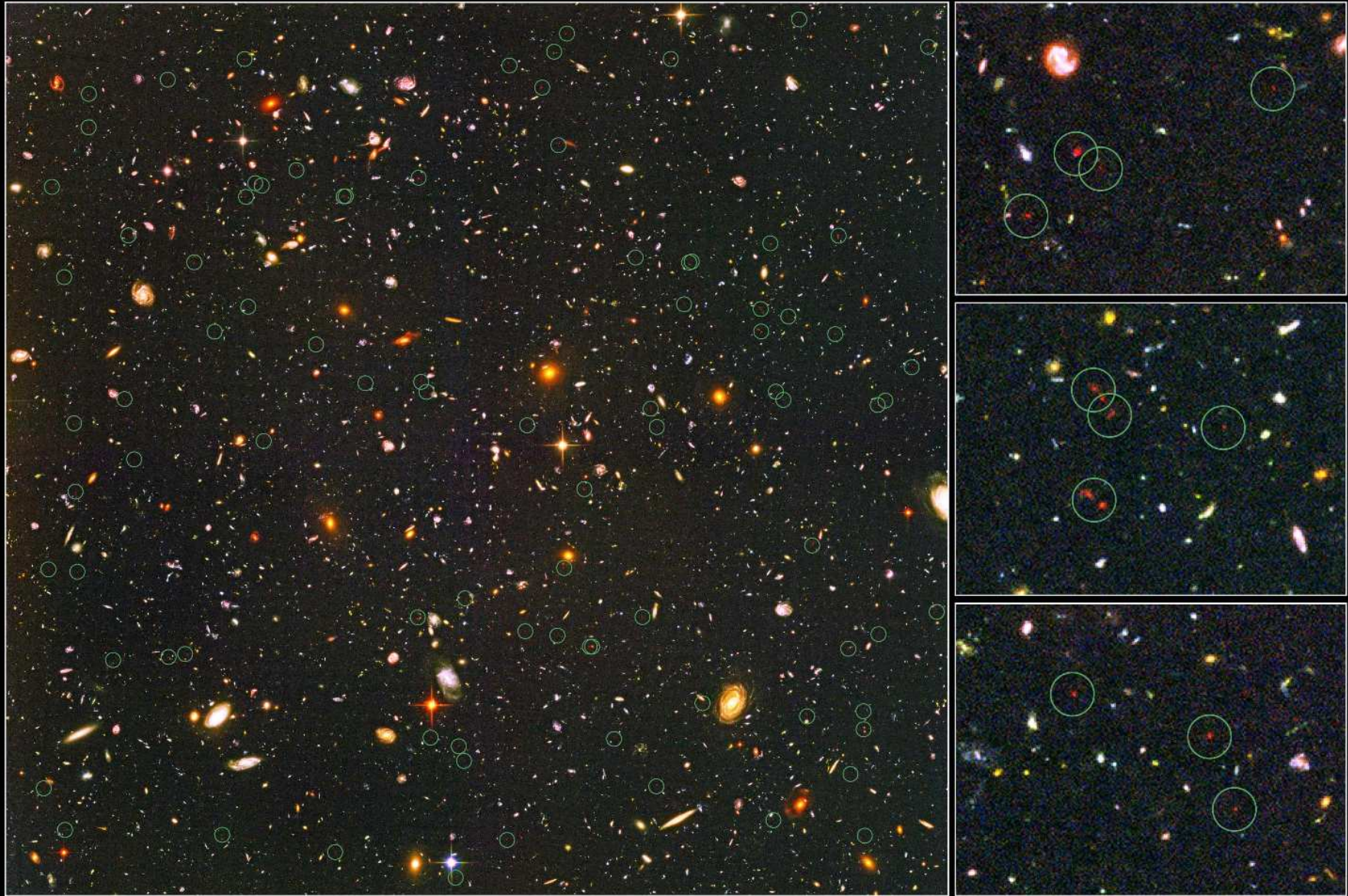




Distant Gravitationally Lensed Galaxy ■ Galaxy Cluster Abell 1689
Hubble Space Telescope ■ ACS/WFC NICMOS



Distant Galaxy Lensed by Cluster Abell 2218
Hubble Space Telescope • WFPC2 • ACS



Distant Galaxies in the Hubble Ultra Deep Field
Hubble Space Telescope • Advanced Camera for Surveys

NASA, ESA, R. Windhorst (Arizona State University) and H. Yan (Spitzer Science Center, Caltech)

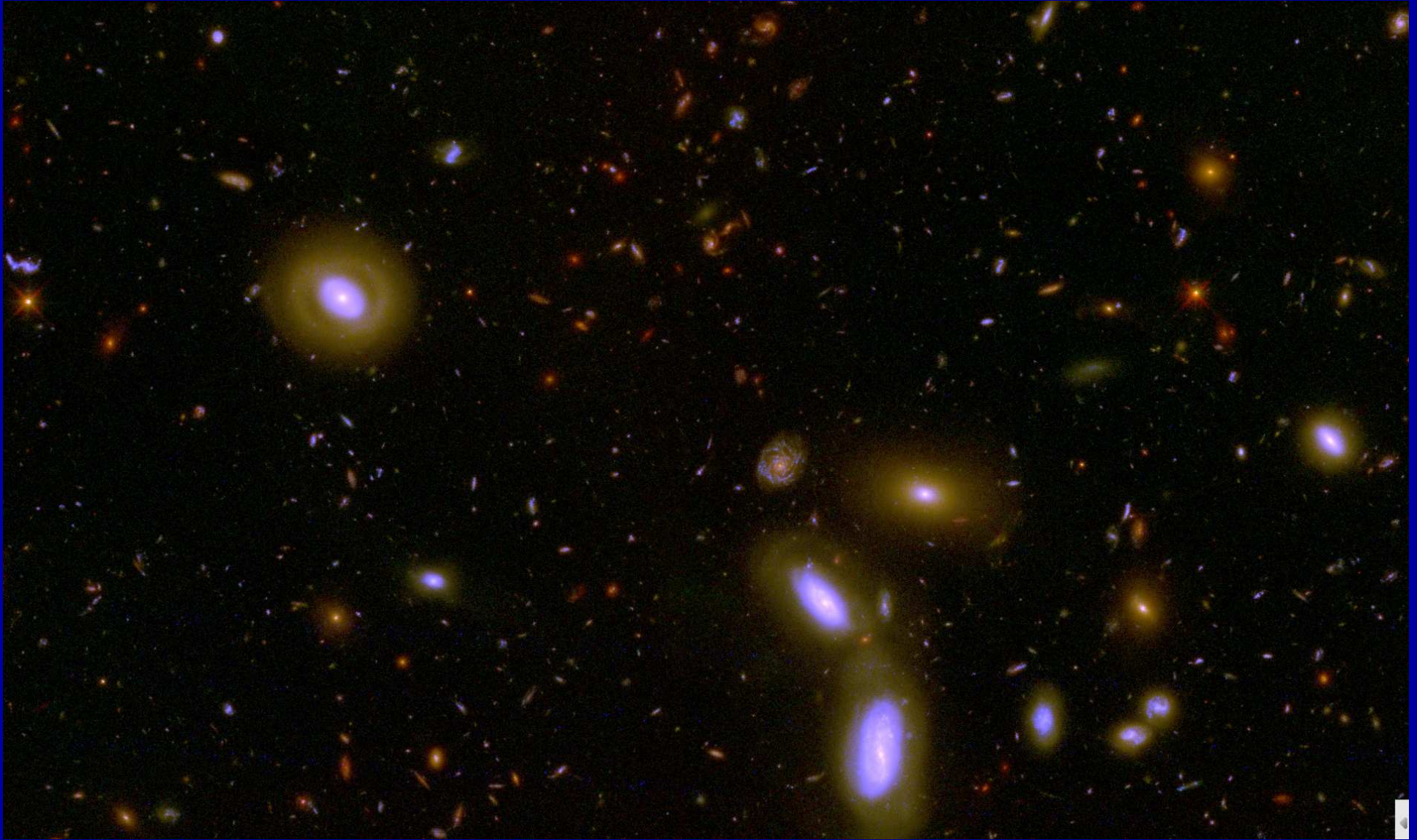
STScI-PRC04-28

A Recent Study of Cosmic Dawn with the Hubble Space Telescope



240 hour image of the Hubble UltraDeep Field: 10^4 galaxies $\lesssim 10^{10} \times$
fainter than unaided eye can see, in area smaller than looking through straw!

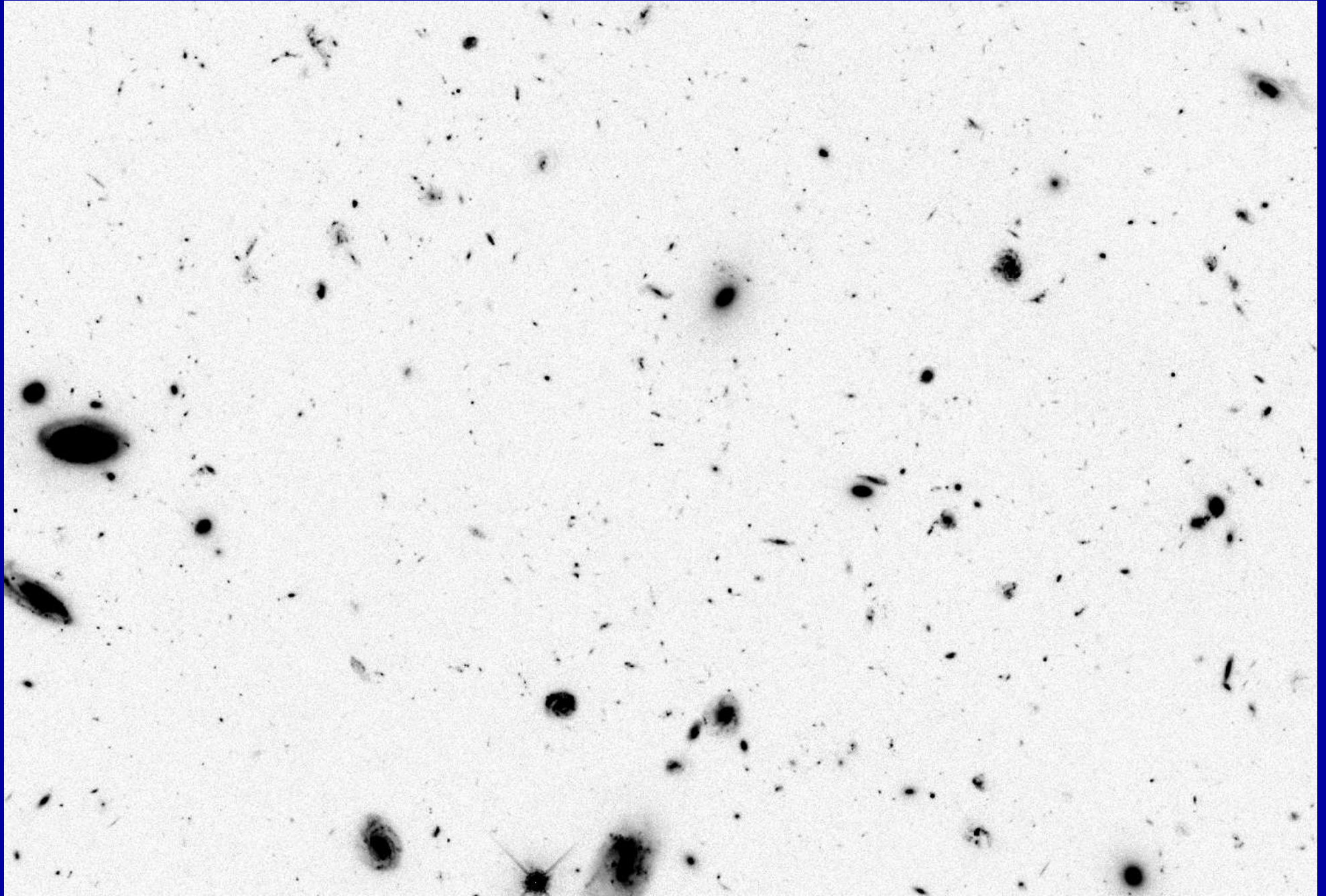
Hubble's new Panchromatic view of the Universe



10 filter color image with the new Hubble Wide Field Camera 3 reaching 10^{-33} W/m²/Hz ($10^9 \times$ fainter than eye) from 0.2–2.0 μ m.

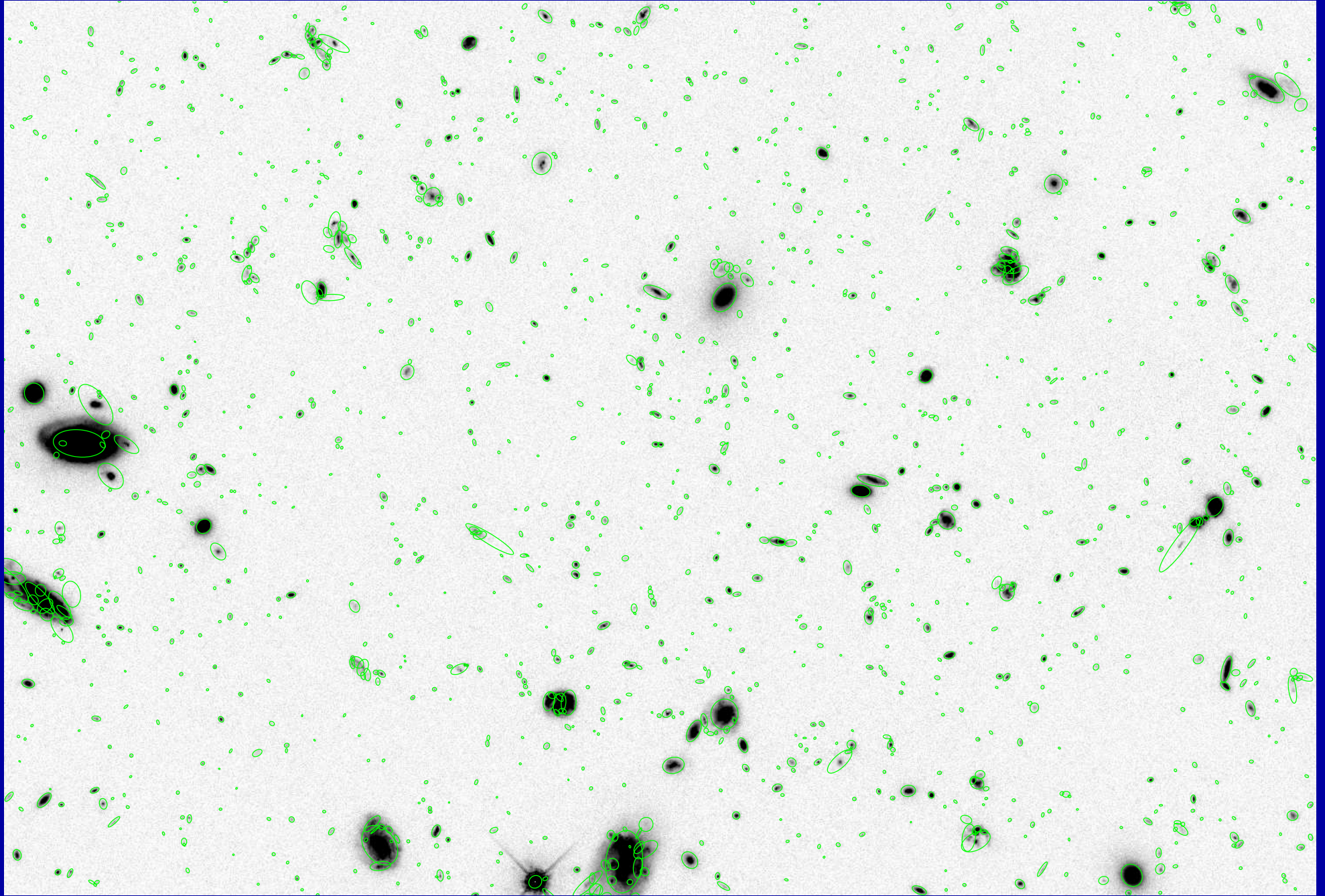
Area is 40 arcmin² (=1/3 of full Moon in diameter), reaching 0.07–0.15" FWHM resolution (=a dime on the Capitol as seen from NYC).

(1b) NASA Hubble Space Telescope image analysis



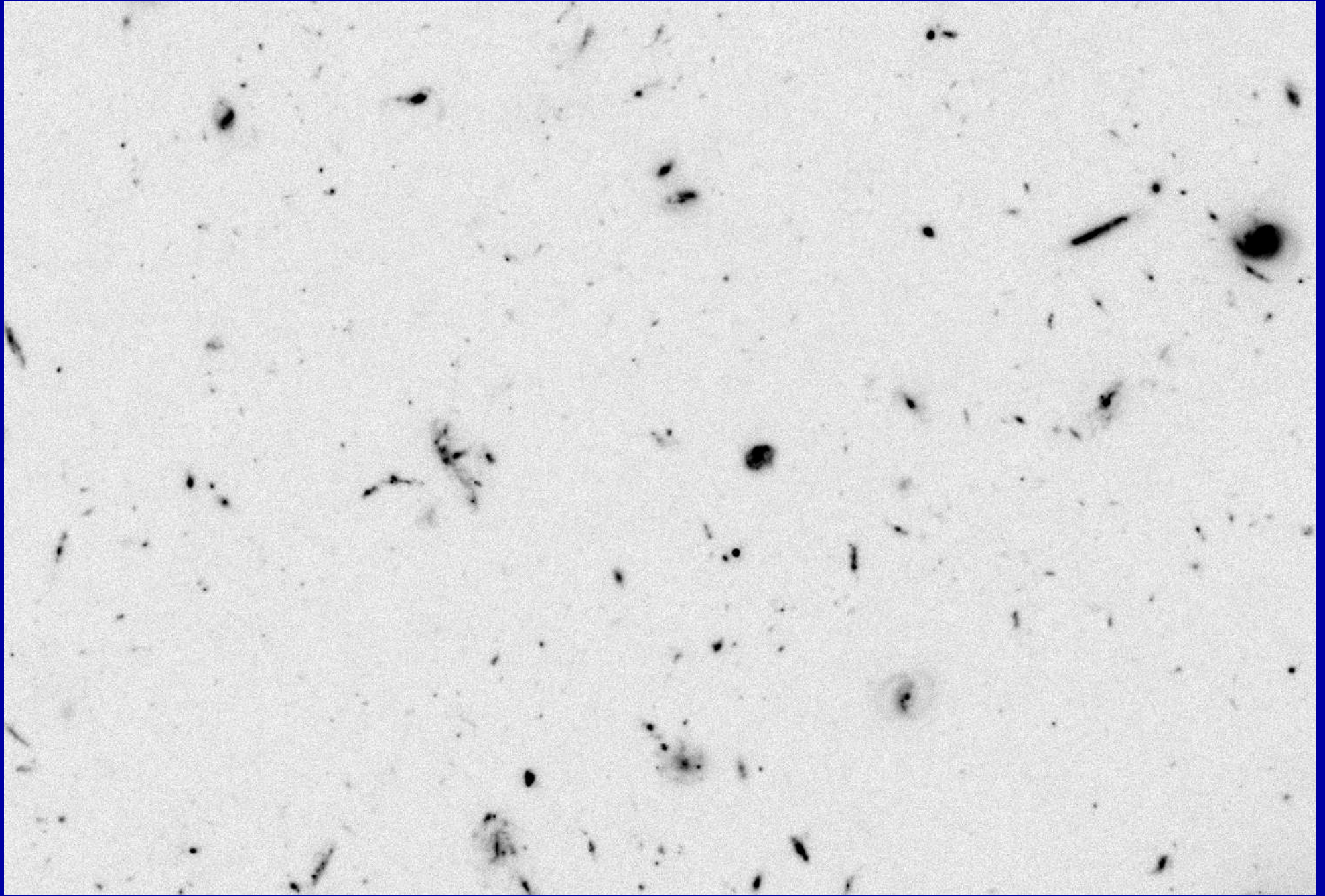
The deepest Hubble images start showing significant galaxy overlap.

(1b) NASA Hubble Space Telescope image analysis



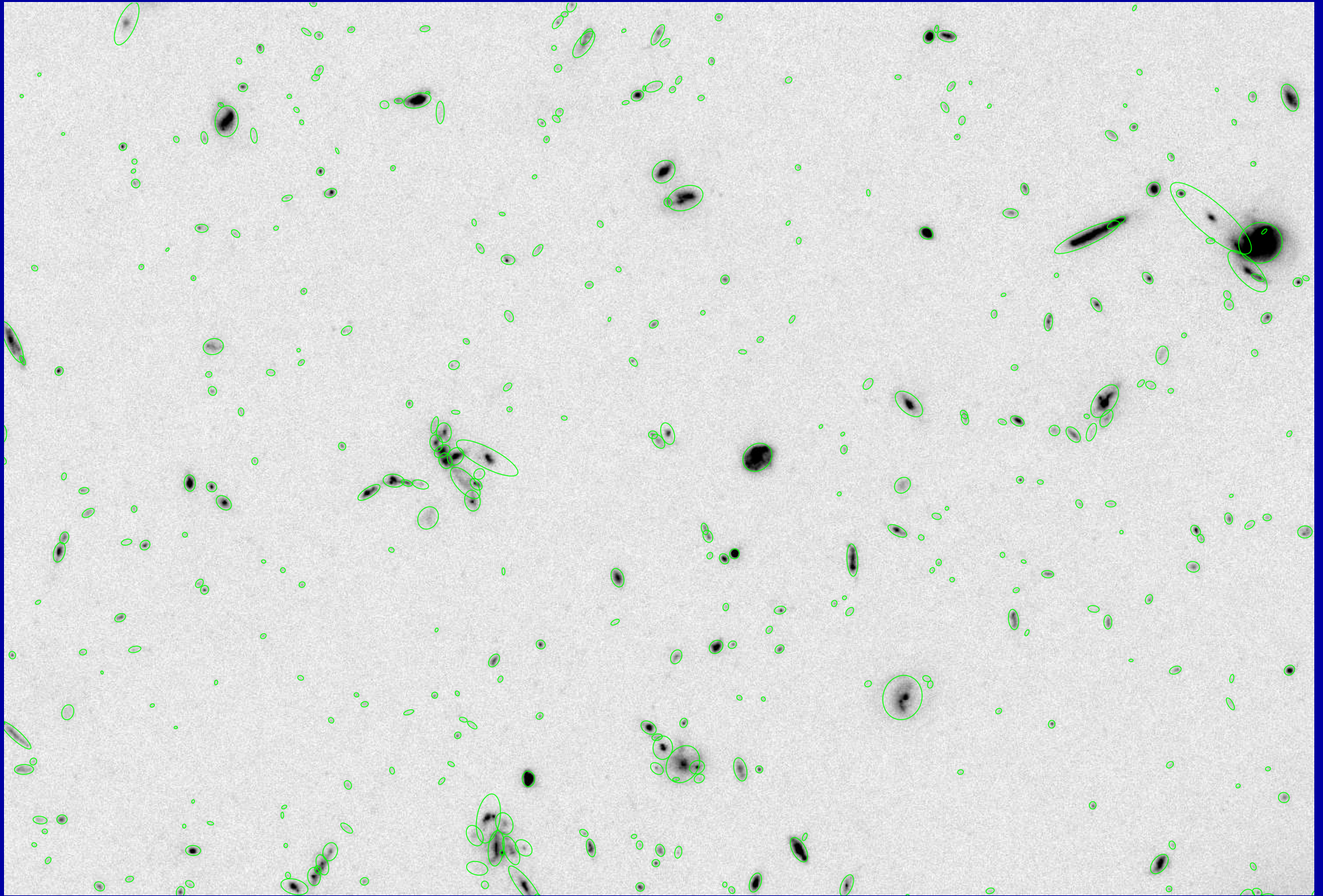
to address crowding, the source-extractor code iteratively deblends the images.

(2a) Object Finding, Detection, Deblending, and Classification method



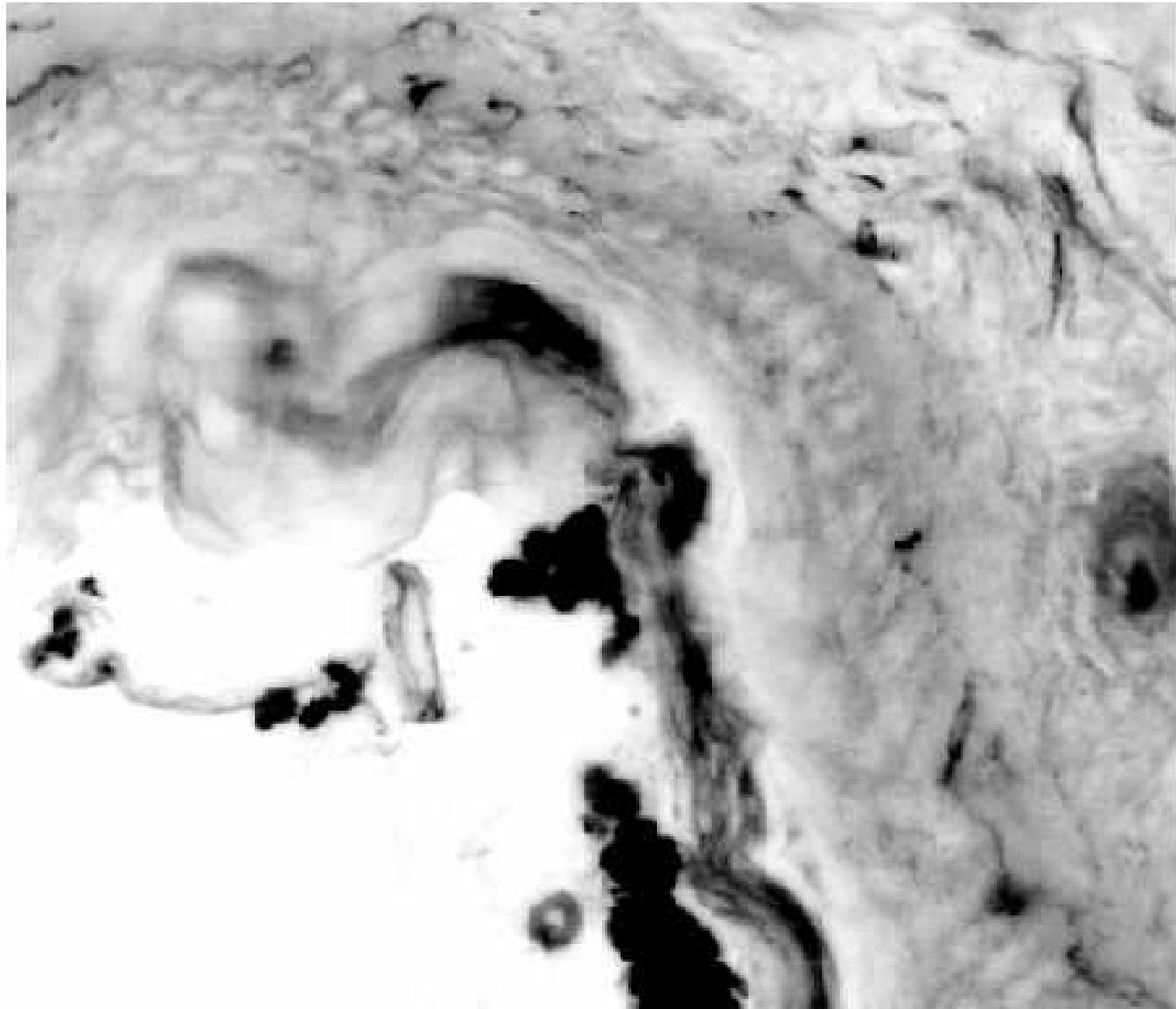
Very complex filamentary galaxies are seen in early stages of formation.

(2a) The Galaxy Detection, Deblending, and Classification method



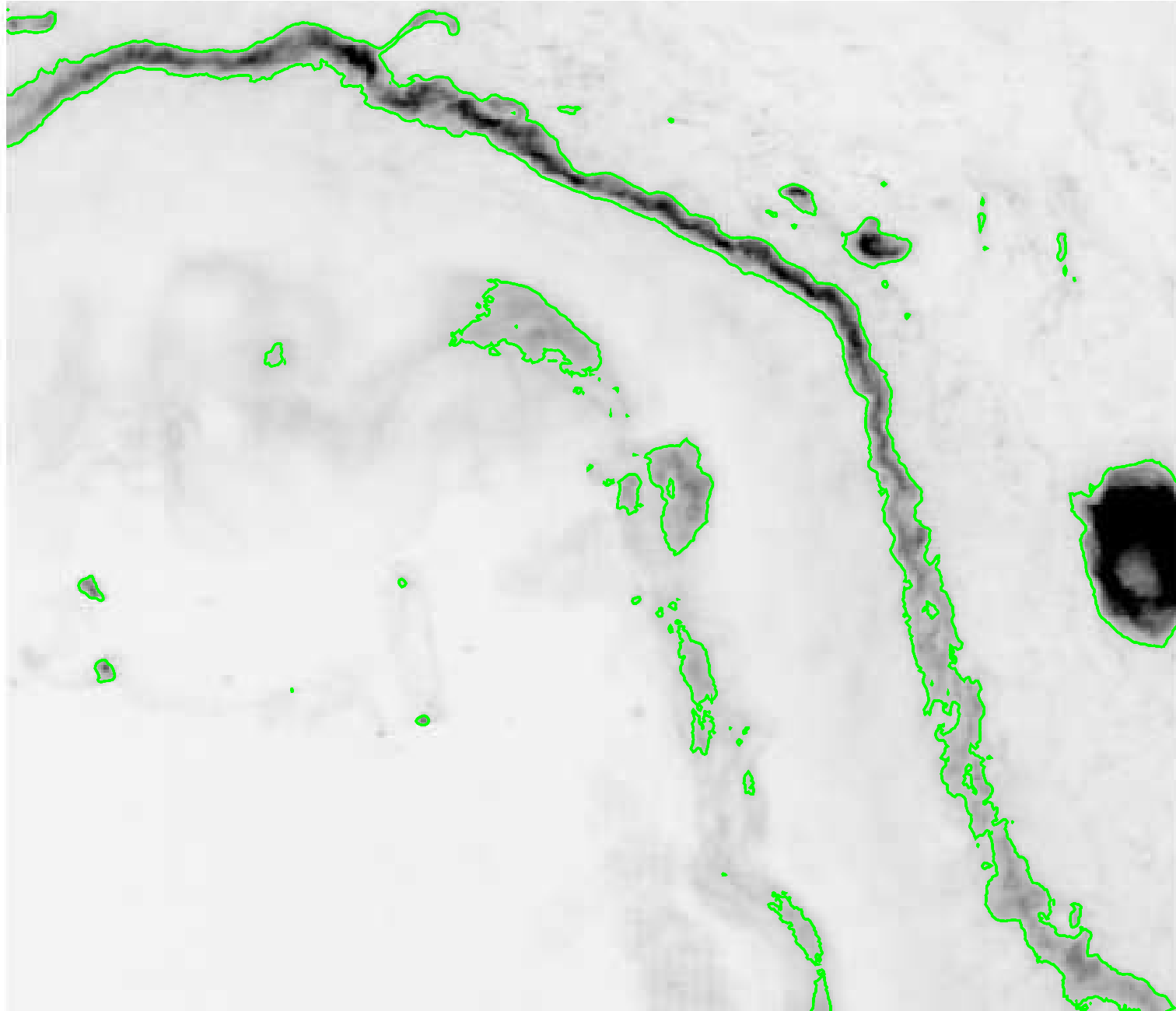
The Source-Extractor software deblends these in chains of elliptical isophotes.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



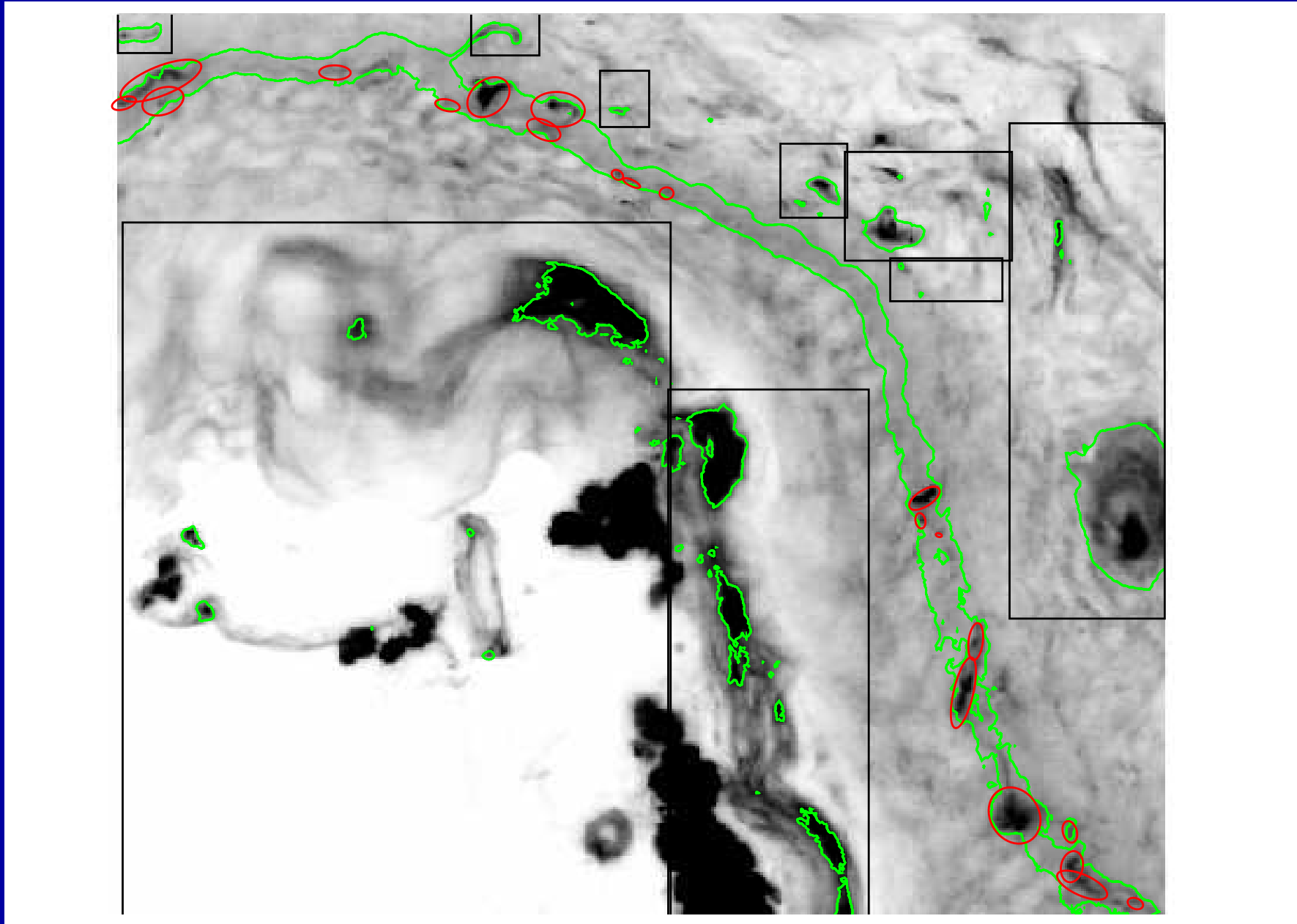
A Stained Confocal Microscope image of a Skin-Biopsy is digitized.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



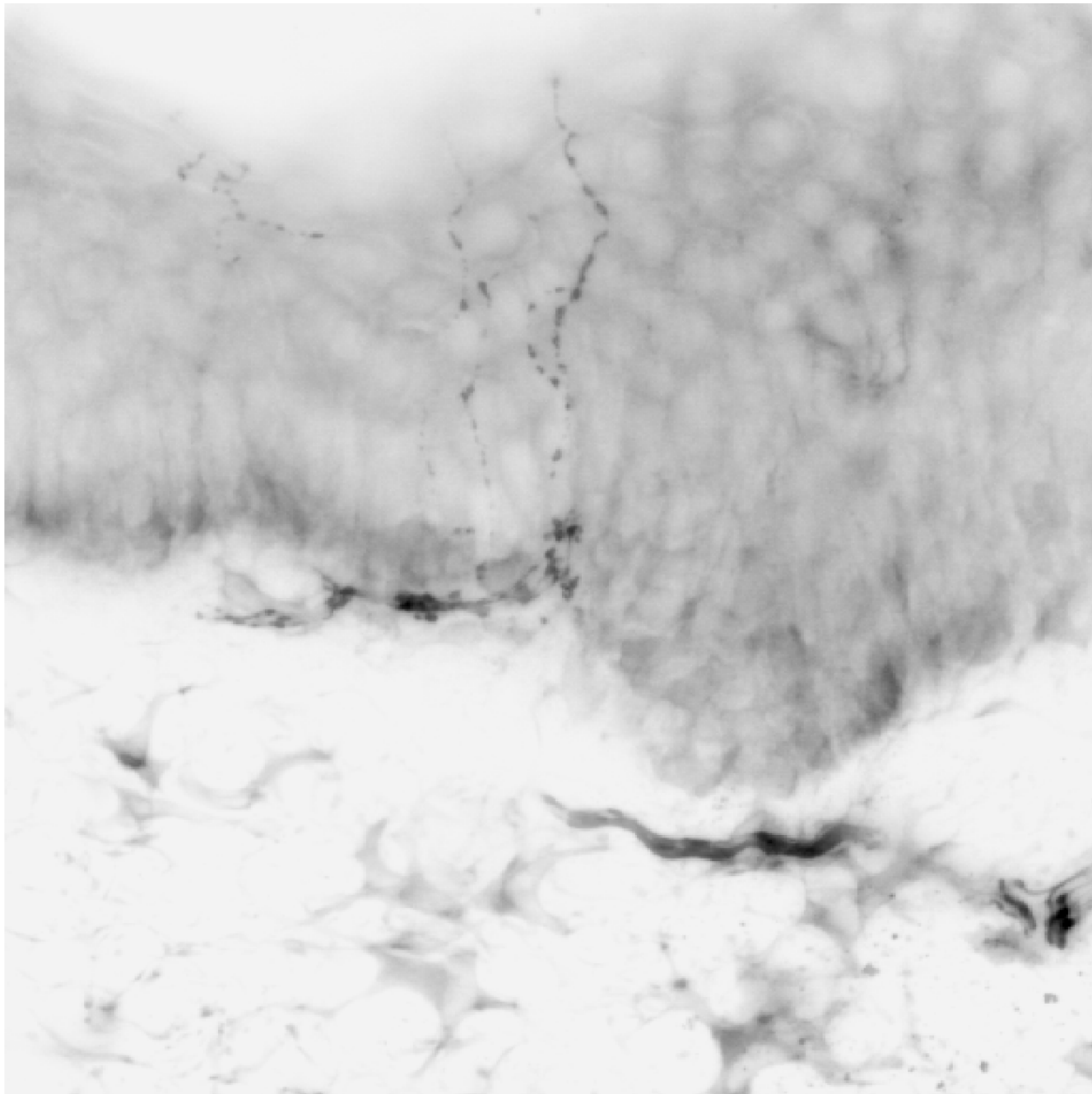
Different staining allows us to define the membrane and other parts of skin.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



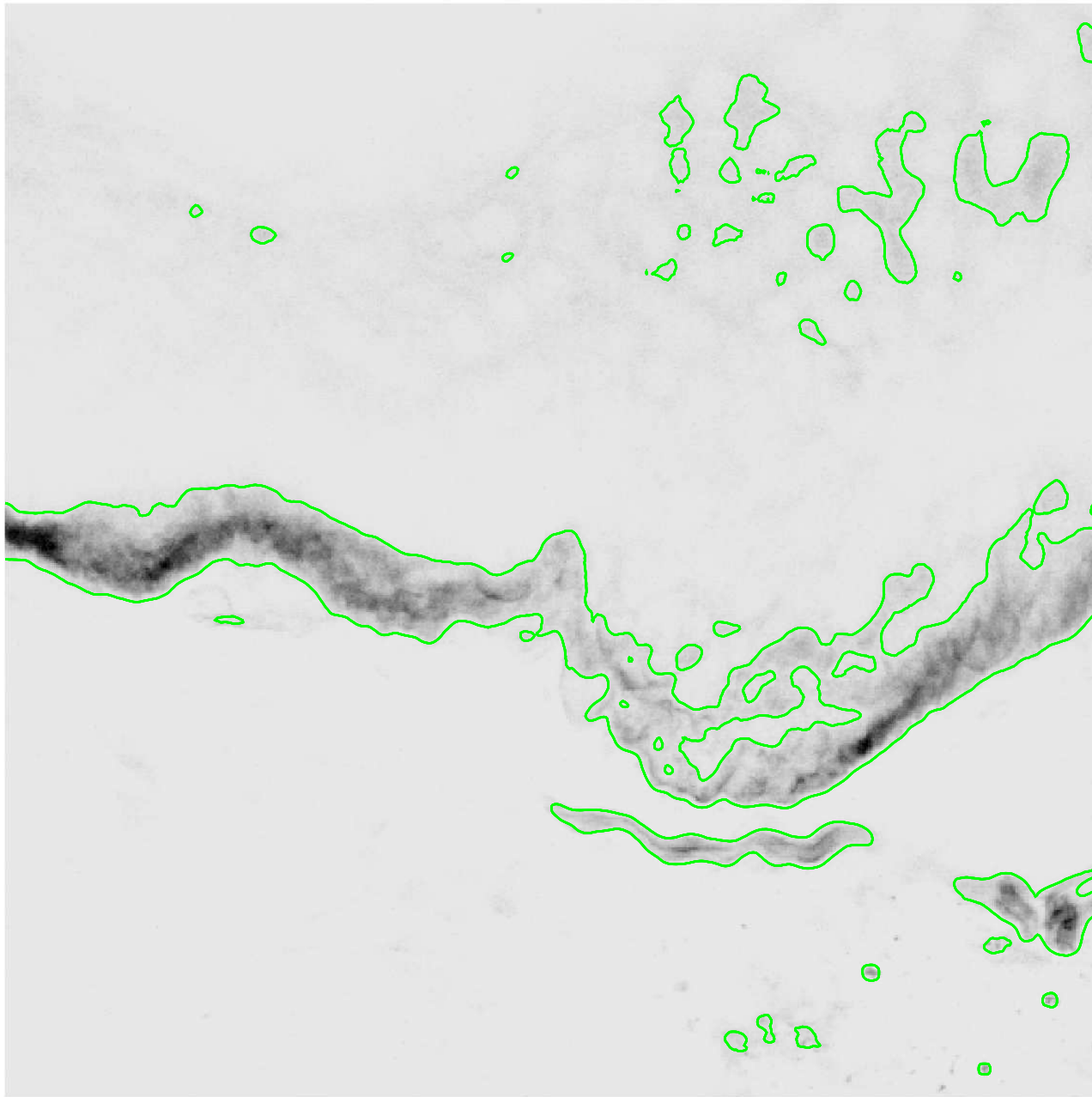
The Source-Extractor finds C-fibers that cross the membrane (in red).
Black boxes: Exclusion zone where Source-Extractor avoids bogus objects.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



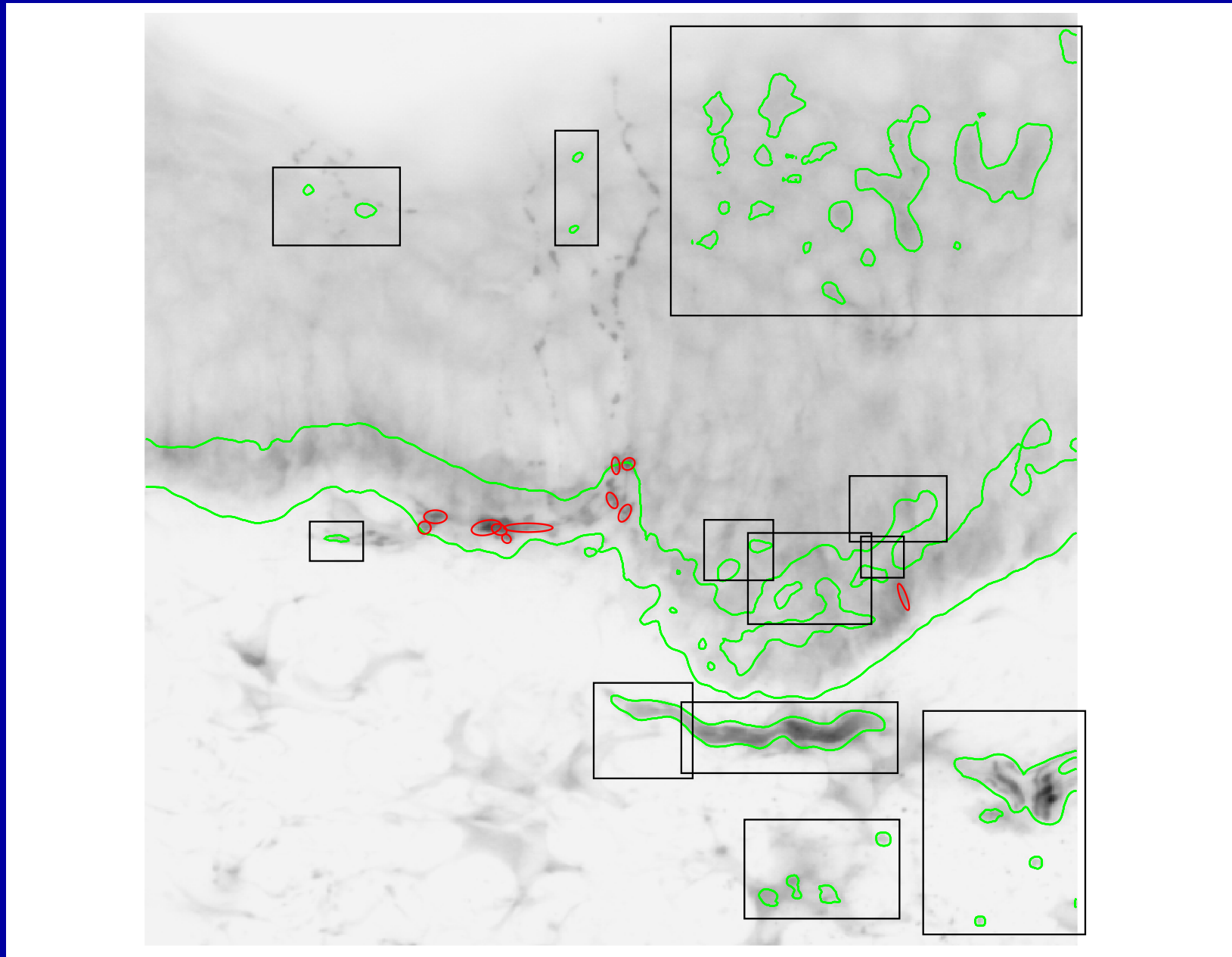
A Stained Confocal Microscope image of a Skin-Biopsy is digitized.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



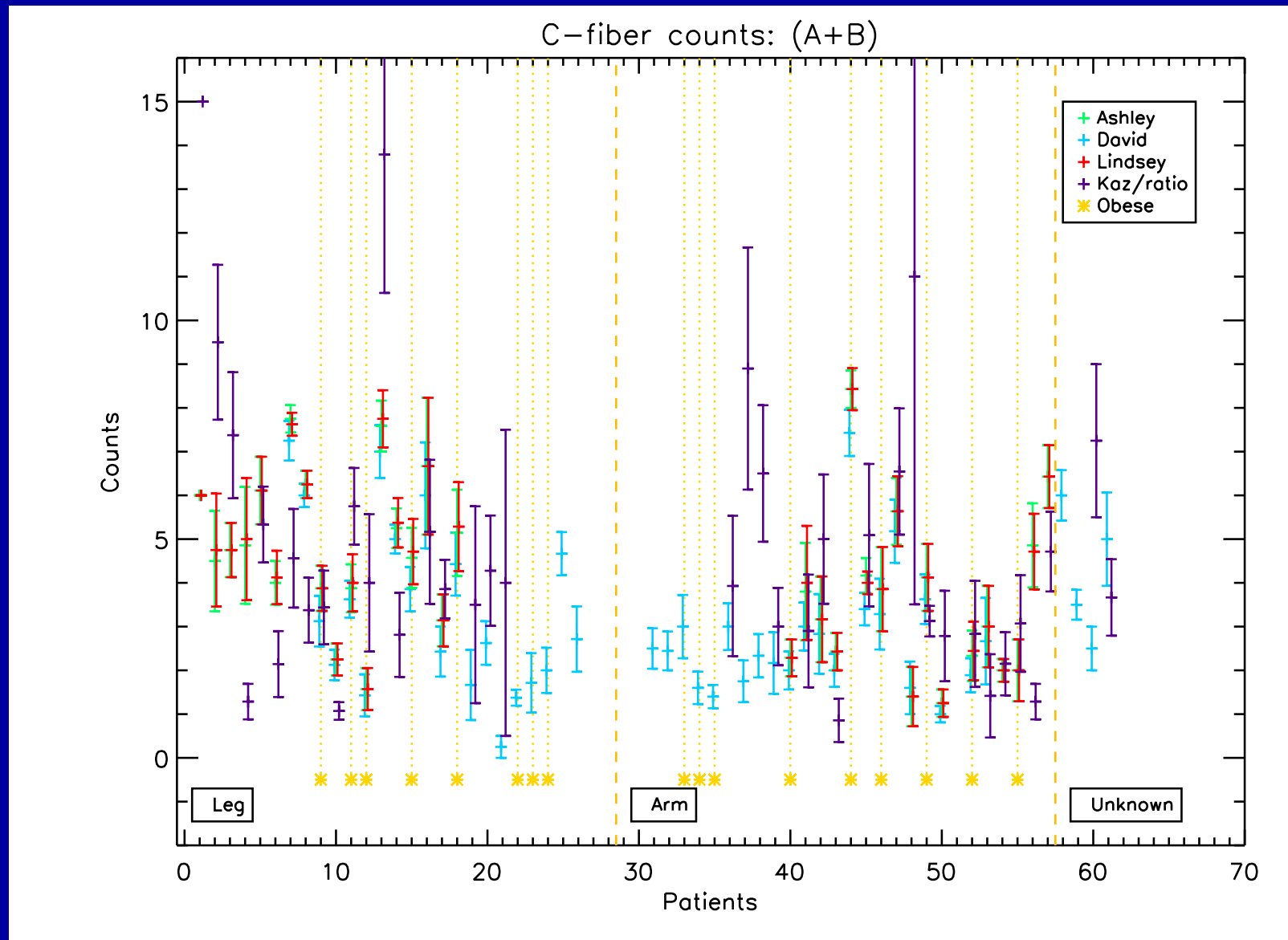
Different staining allows us to define the membrane and other parts of skin.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



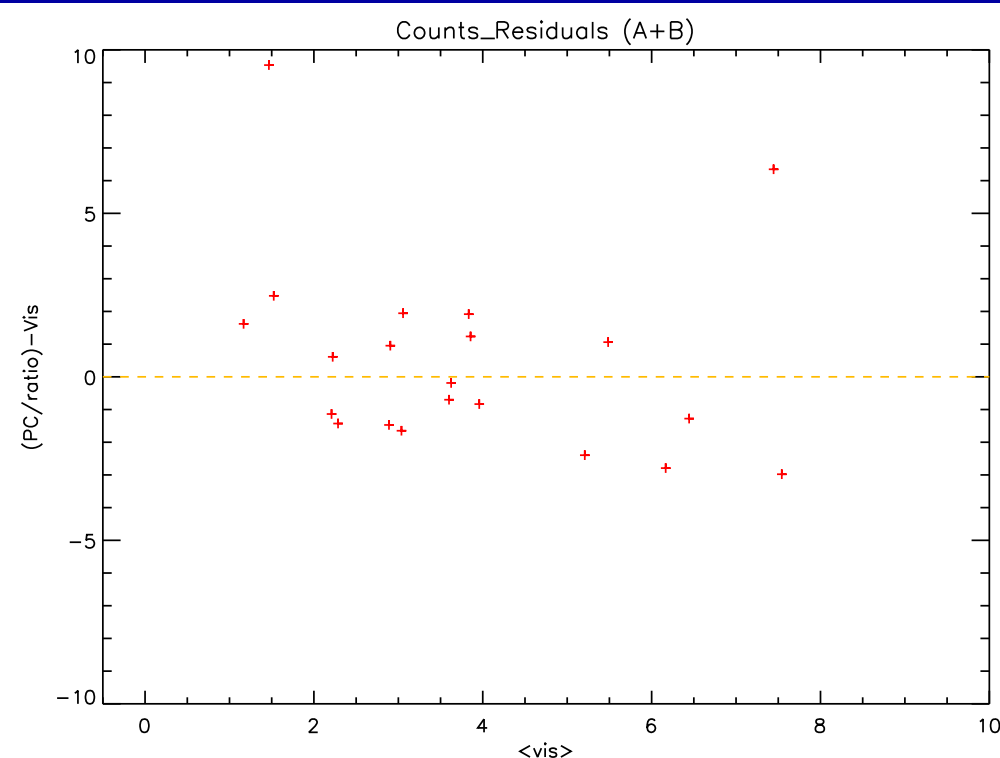
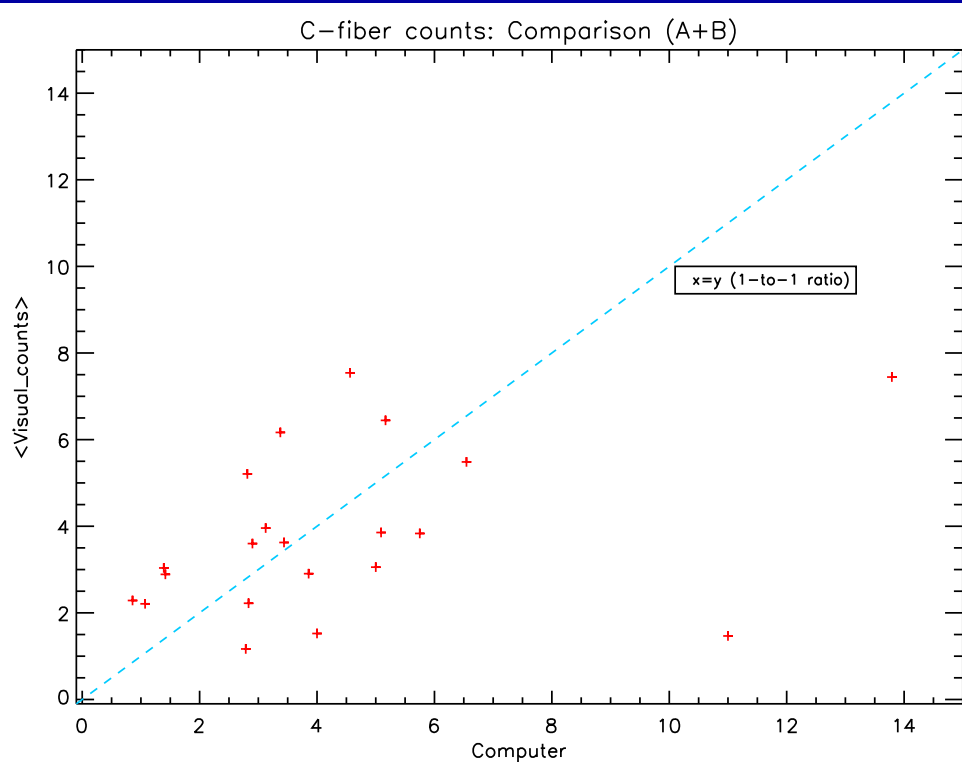
The Source-Extractor finds C-fibers that cross the membrane (in red).
Black boxes: Exclusion zone where Source-Extractor avoids bogus objects.

(2b) Application to C-fiber Skin-biopsies



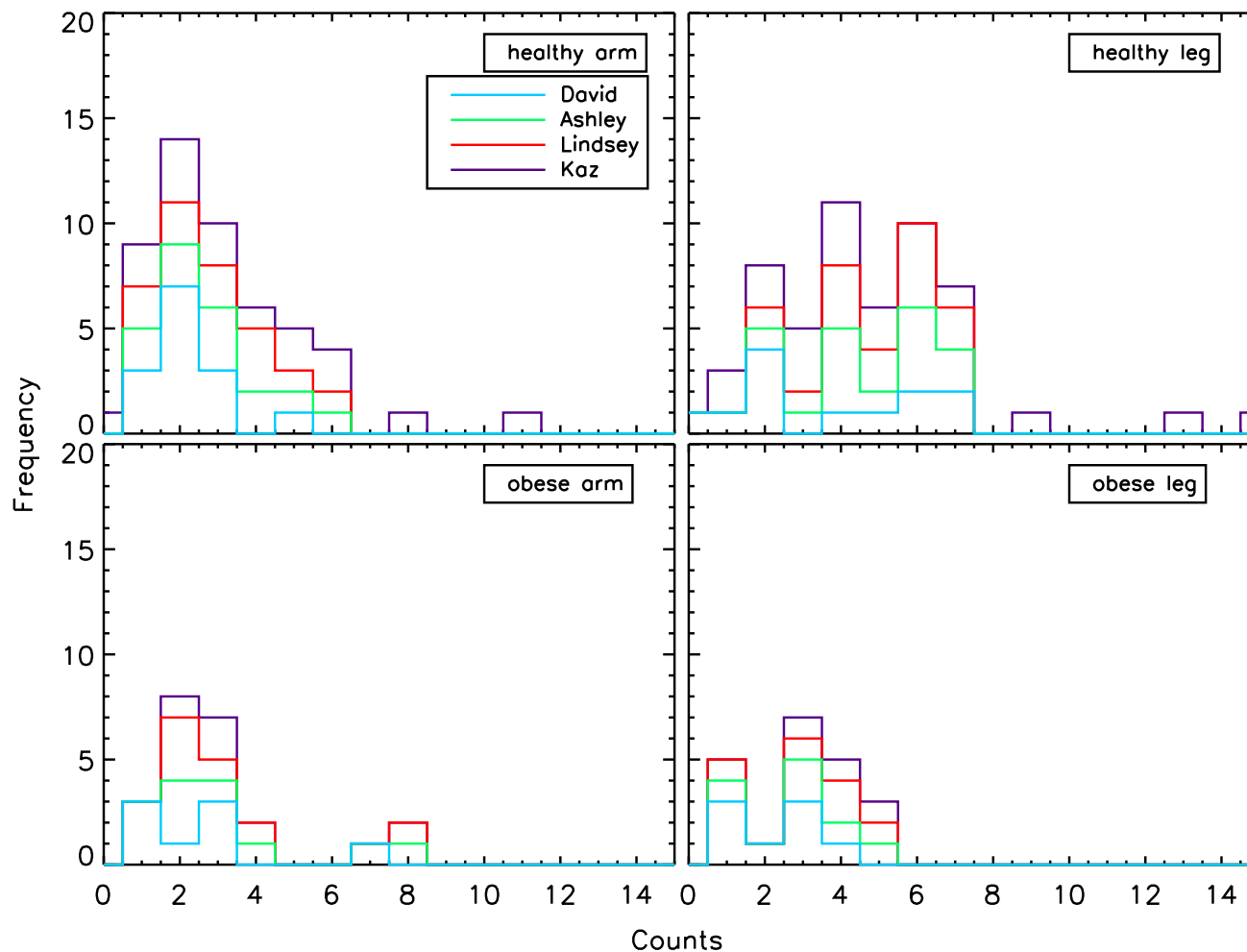
- Machine classification (black) follows average of 3 independent visual classifiers (different colors) quite well.
- Obese patients are marked in yellow, and have often lower C-fiber count.

(2b) Application to C-fiber Skin-biopsies



- Machine count correlates well with visual counts, with a few catastrophic failures where software breaks a C-fiber up in too many pieces.
- Average fragmentation is calibrated from expert visual classifiers.
- Catastrophic failures are corrected by hand — working on training the software to recognize and correct those as well.
- Dispersion between machine and visual classifiers is $\sigma \simeq \pm 1.5$ count.

(2b) Application to C-fiber Skin-biopsies



- Cumulative counts of both the **machine** and the **visual** classifiers show different distributions for normal and obese patients.
- Obese patients have in general lower C-fiber counts, although the differences between the two populations are not always huge.

(3) Conclusions

(1) Summary of the NASA Hubble Space Telescope connection

- (1a) Astronomers have no choice but to study 99.99999999999999999999999999990 % of the universe with clever software.
- (1b) NASA Hubble Space Telescope image analysis software can detect, deblend and classify faint galaxy or faint star images.
- (1c) Software is very general — doesn't care what it counts or classifies!

(2) Can use software to identify potential markers for Diabetes Type 2 in early stage:

(2a) Galaxy Detection, Deblending, and Classification software is fairly robust and general, minimizing human interaction.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese patients allows us to quantitatively distinguish healthy from sick patients. Dispersion with visual classification is $\sigma \simeq \pm 1.5$ C-fiber count.

(2c) Much larger samples are needed to better cover sub-samples of: healthy, obese, adult, child, and rare classes. This work is in progress.

REFERENCES

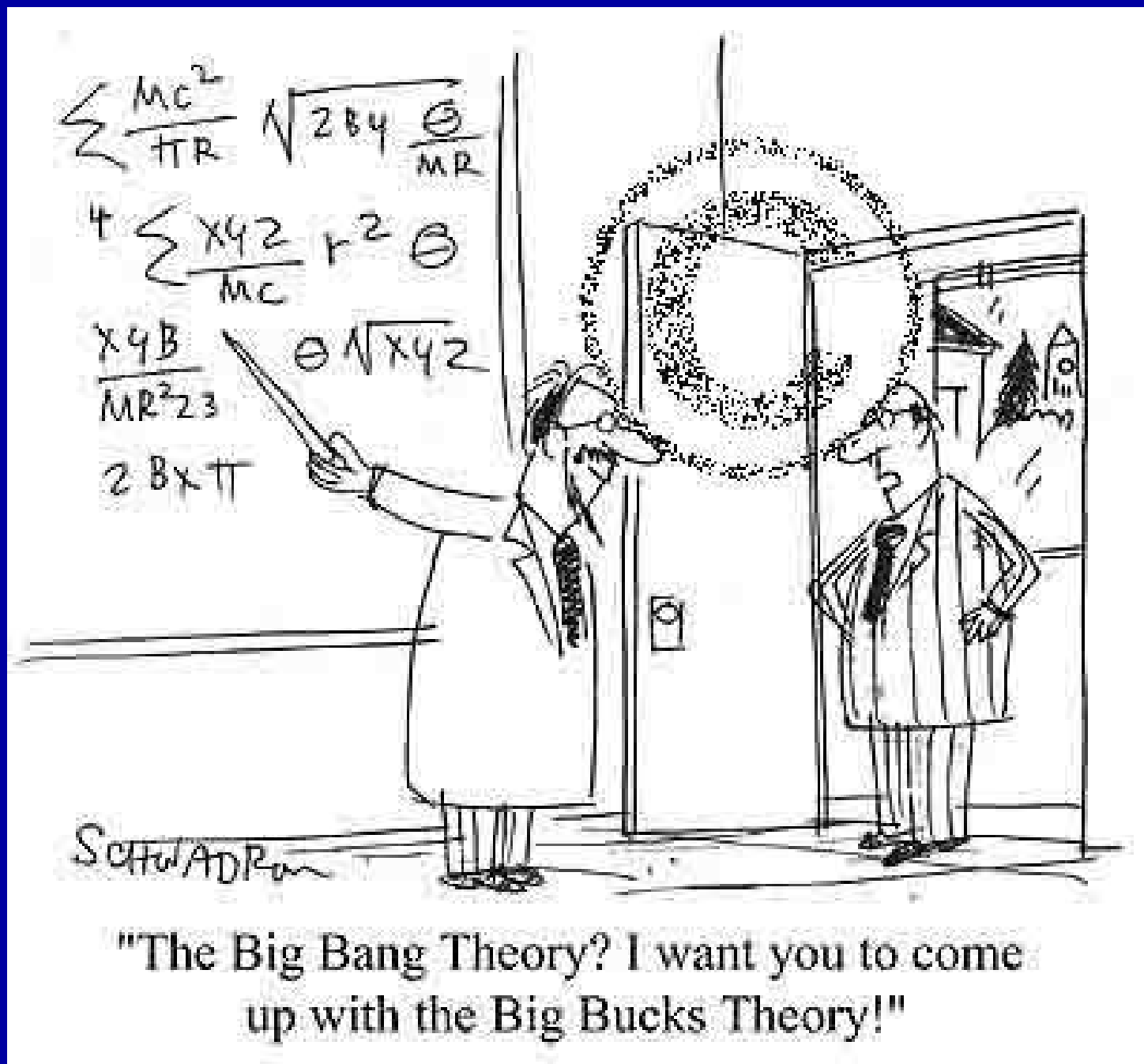
<http://hubblesite.org/newscenter/2010/01/>

Casano, A. R., Brower, J. B., Baluch, D. P., Windhorst, R. A., Targovnik, J. H., & Herman, R. M. 2009, *International Journal of Obesity*, submitted (Jul. 2009) [<http://www.nature.com/ijo/index.html>] “Small Sensory Nerve Fiber Neuropathy in Morbidly Obese Subjects: Reduced Intra Epidermal Nerve Fiber Density”

Herman, R. M., Brower, J. B., Stoddard, D. G., Casano, A. R., Targovnik, J. H., Herman, J. H., & Tearse, P. 2006, *International Journal of Obesity*, 31, 226–235 [www.nature.com/ijo/journal/v31/n2/full/0803418a.html] “Prevalence of somatic small fiber neuropathy in obesity ”

Tamura, K., Mager, V. A., Burnett, L. A., Olson, J. H., Brower, J. B., Casano, A. R., Baluch, D. P., Targovnik, J. H., Windhorst, R. A., & Herman, R. M. 2009, *Journal of Neuroscience Methods*, 185, pg. 325–337 [<http://ees.elsevier.com/jneumeth/>] “Semi-Automated Method of Analysis of Small Sensory Nerve Fibers in Human Skin-Biopsies”

SPARE CHARTS



Some cosmologists at ASU decided to step out of the box ...
(although it is hard to please the dean ...)

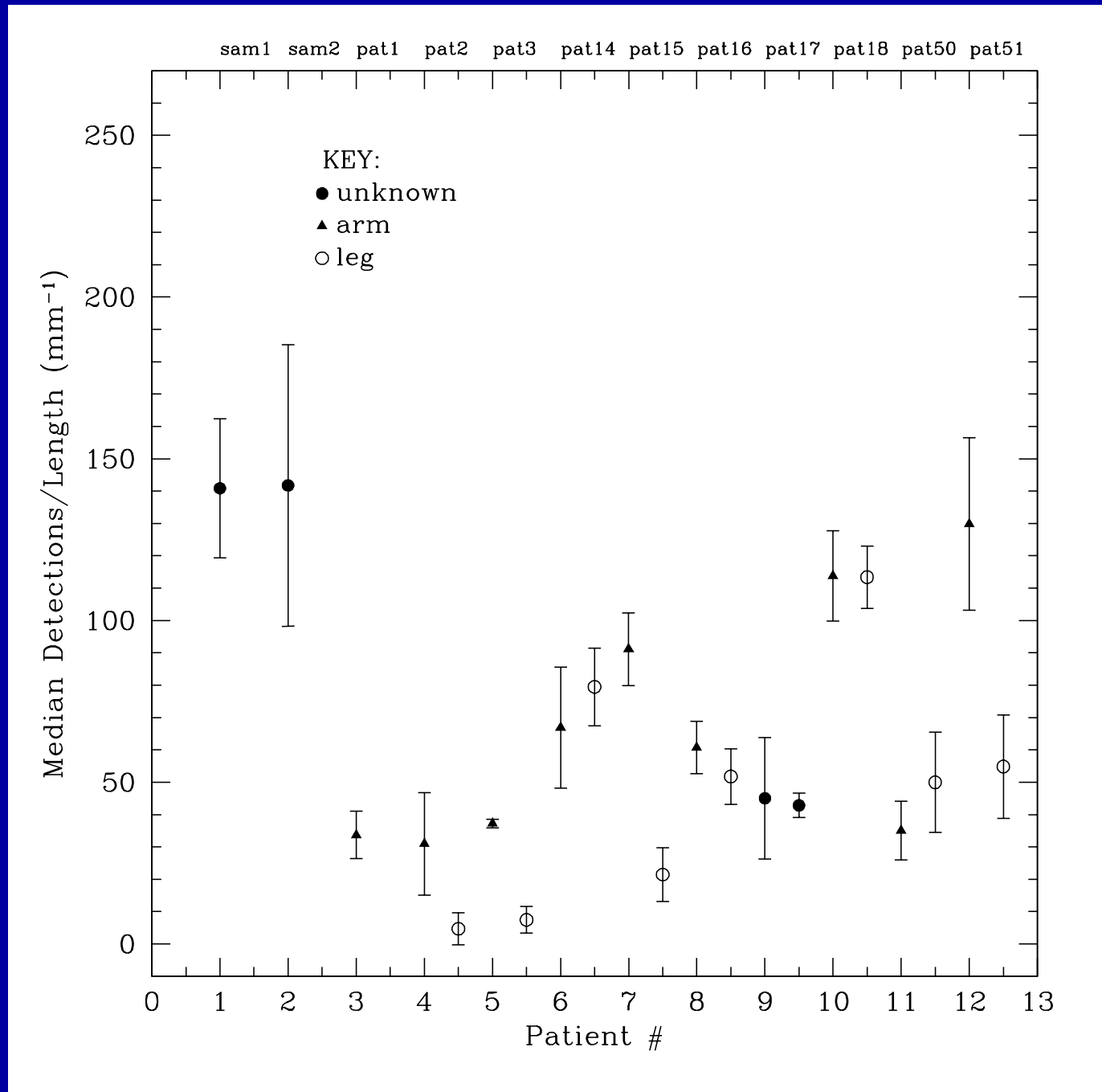
(1) The NASA Hubble Space Telescope connection

- [illegible]

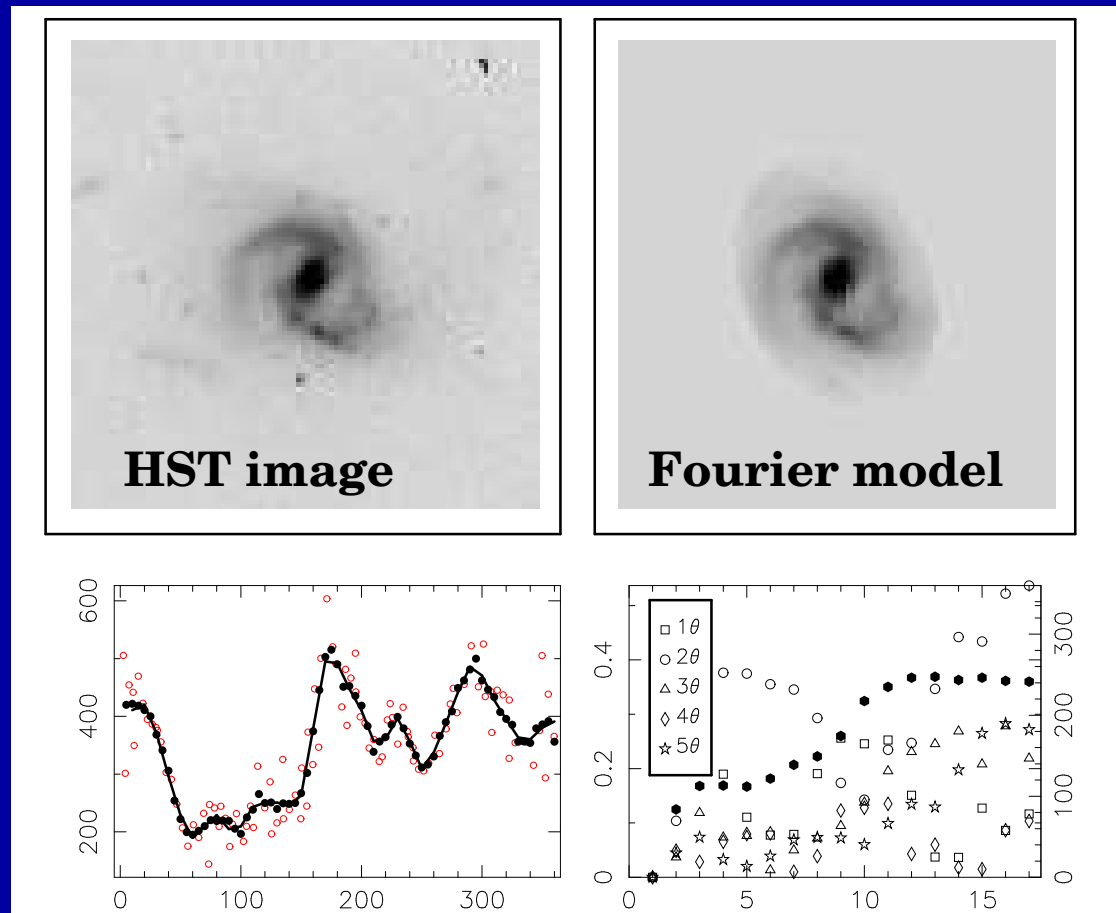
⇒ Astronomers must be smart about using software to analyze what they see in Hubble images — we have absolutely no alternatives!.

This project of medical applications of Hubble galaxy classification software builds on a 40 year history of astronomical software development.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



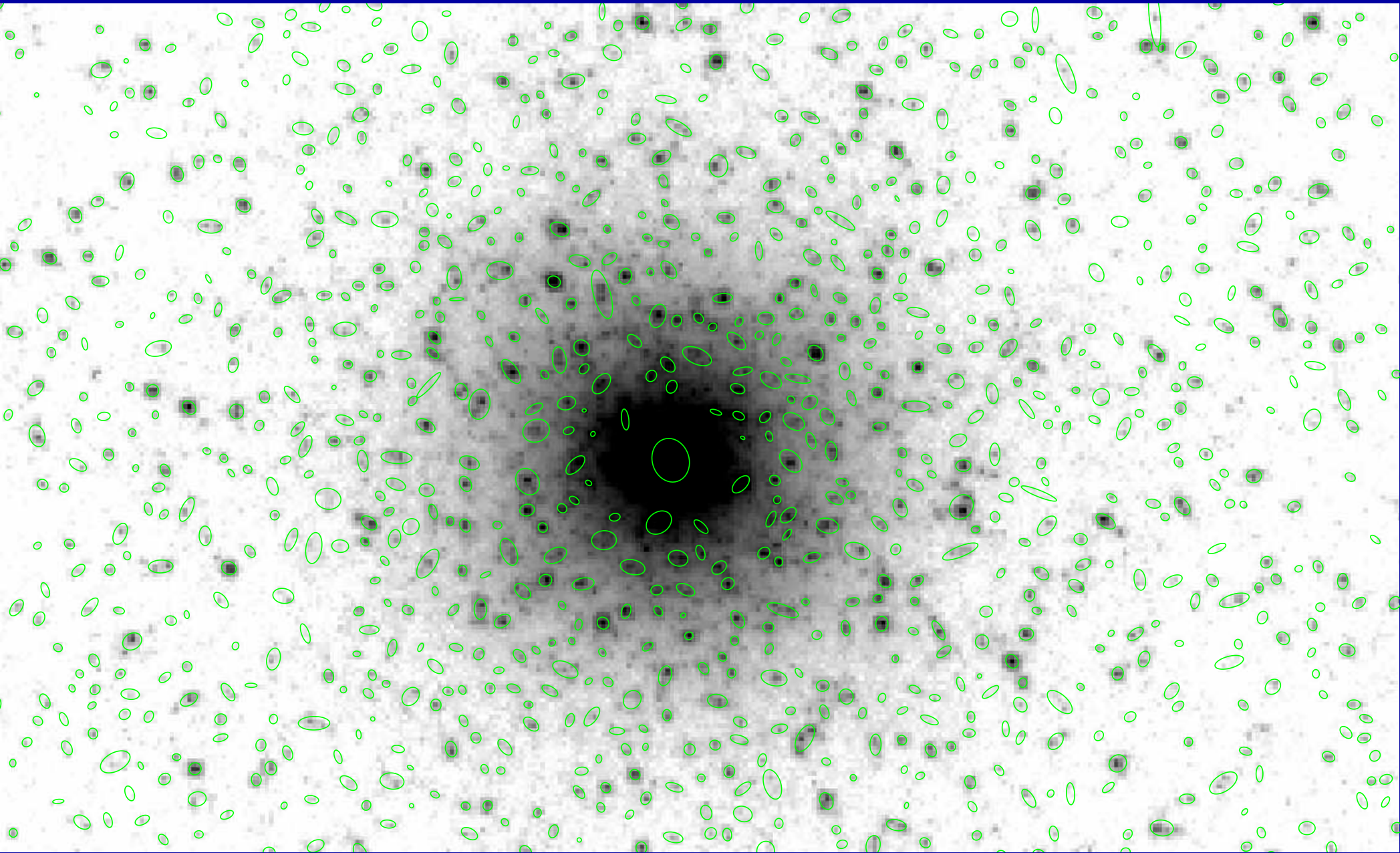
Number of C-fibers crossing membrane is directly counted for each patient.



Fourier Decomposition is a robust way to measure galaxy morphology and structure in a quantitative way (Odewahn et al. 2002):

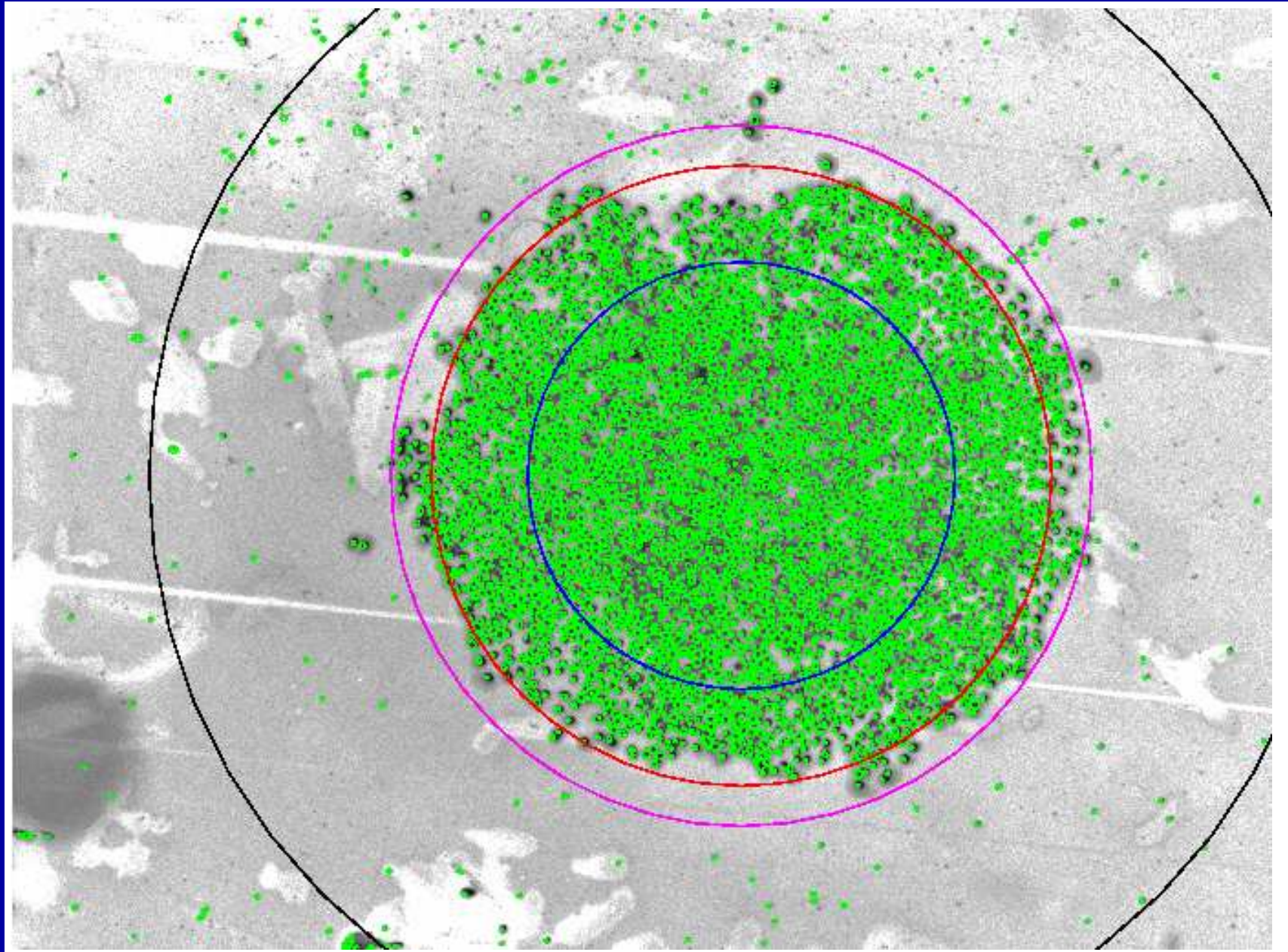
- (1) Fourier series are made in successive concentric annuli.
- (2) Even Fourier components indicate symmetric parts (arms, rings, bars).
- (3) Odd Fourier components indicate asymmetric parts (lopsidedness).
- (4) JWST can measure the evolution of each feature/class directly.

(2c) The Galaxy Detection, Deblending, and Classification method



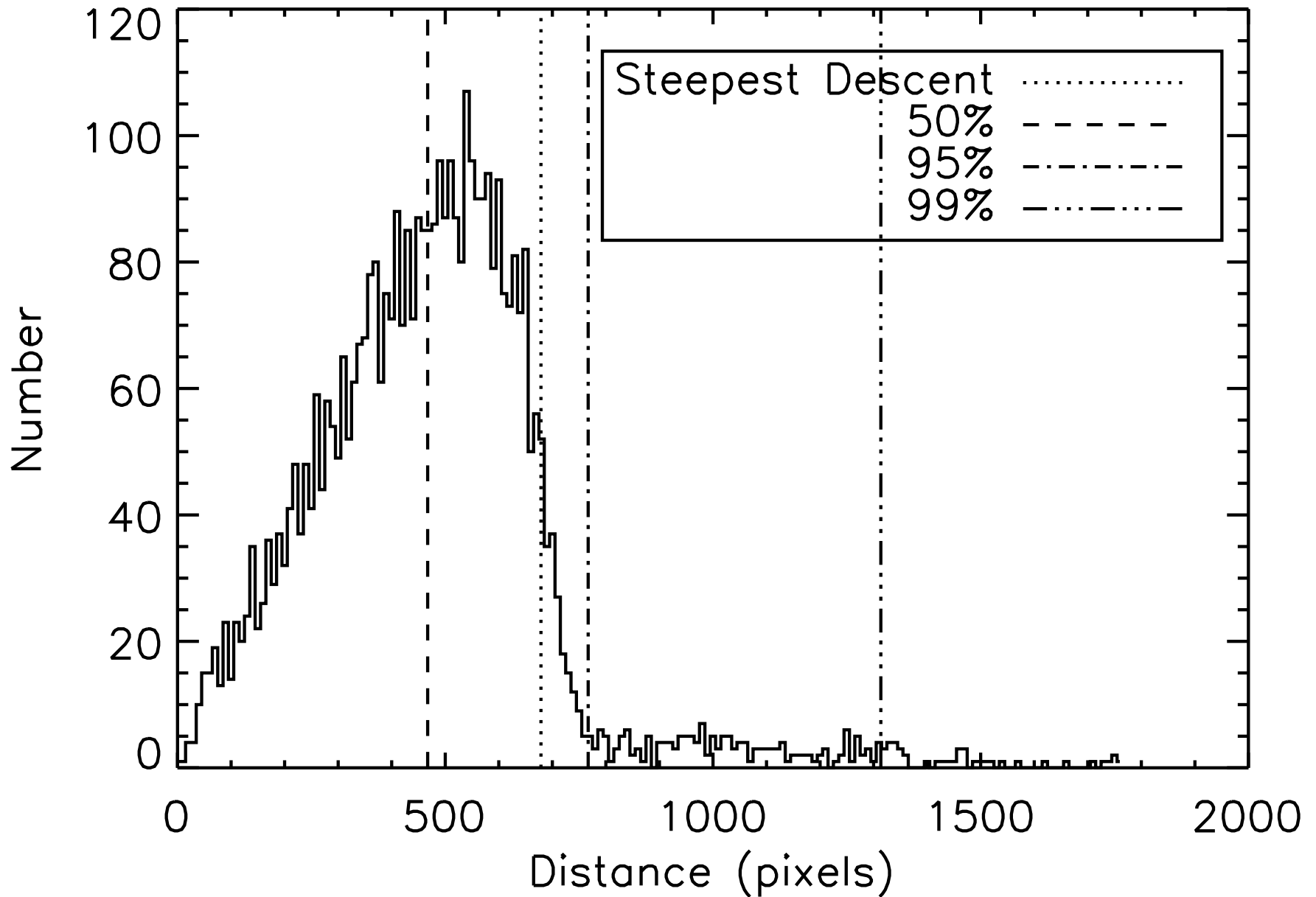
The Source-Extractor algorithm deblends a rich globular cluster of 10^5 stars.
(Only several 10^3 brightest stars plotted, but object overlap is allowed).

(2c) Application of Hubble Software to Study Spreading of Cancer Cells



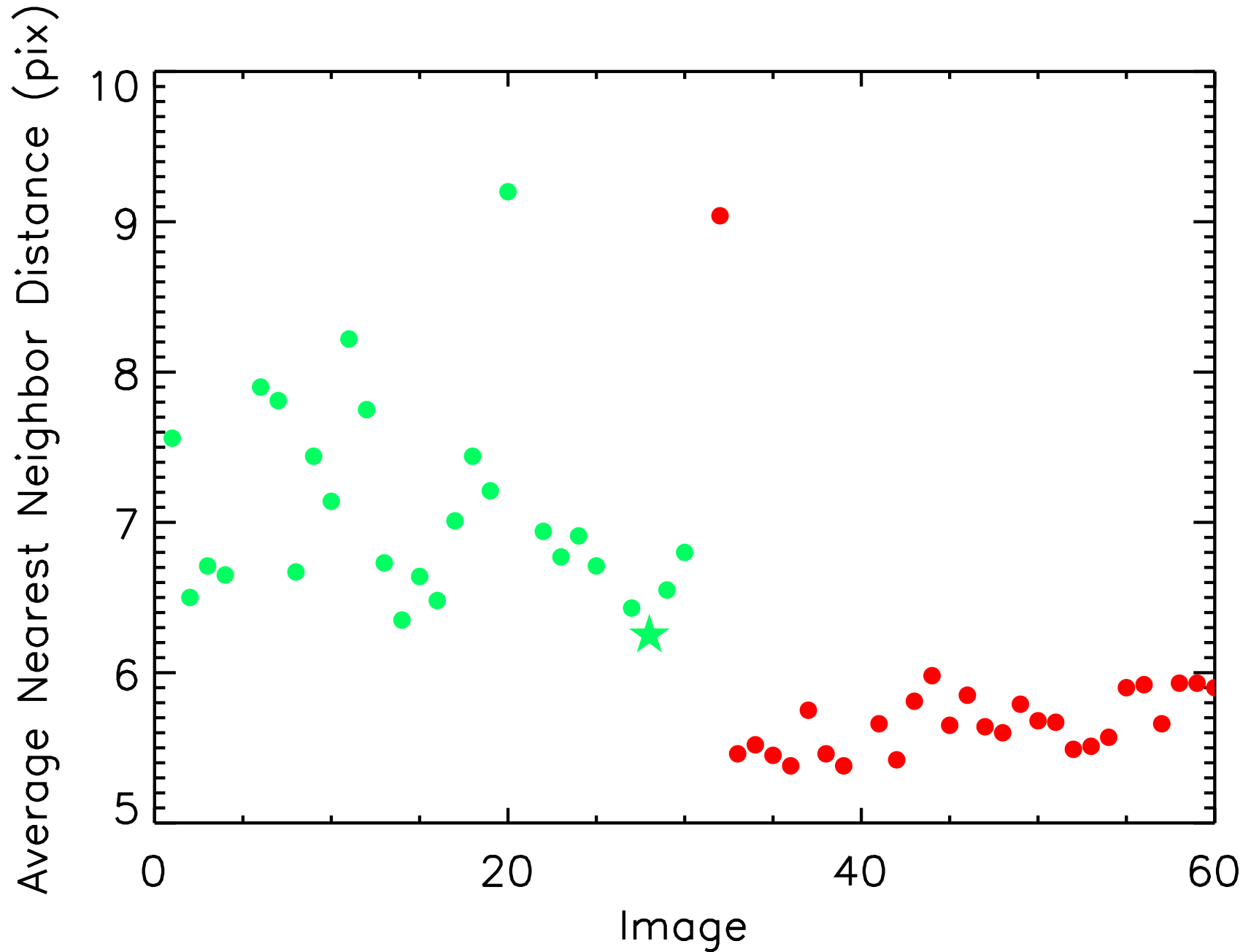
Source-Extractor directly counts number of cells in a cancer biopsy image.
(Collaboration with Dr. M. Berens et al. at TGEN, R. Ryan at ASU.)

(2c) Application of Hubble Software to Study Spreading of Cancer Cells



The distribution and spreading of cancer cells is measured quantitatively.

(2c) Application of Hubble Software to Study Spreading of Cancer Cells



Nearest neighbor analysis bifurcates the spread and non-spread cancer biopsies.